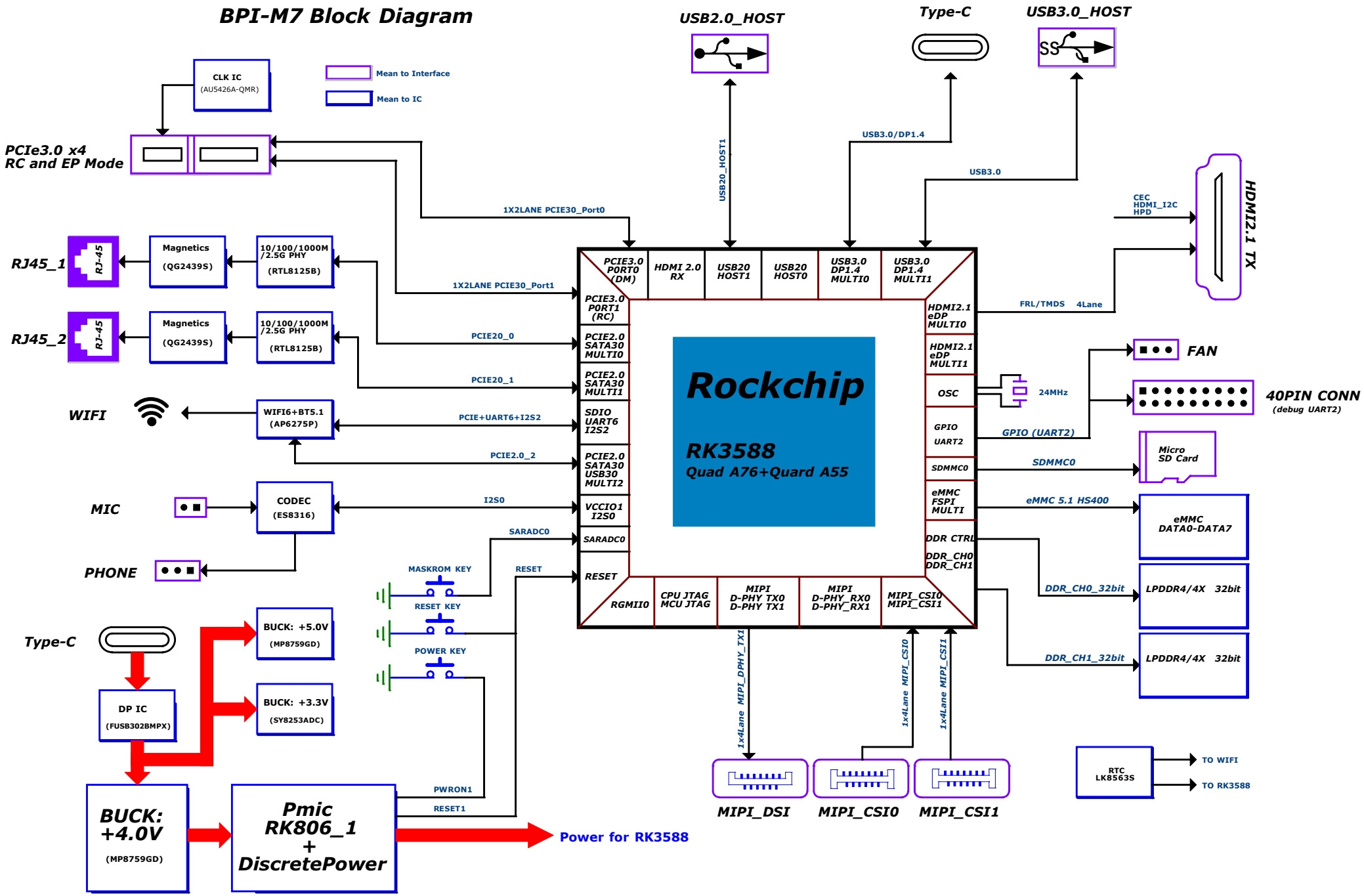
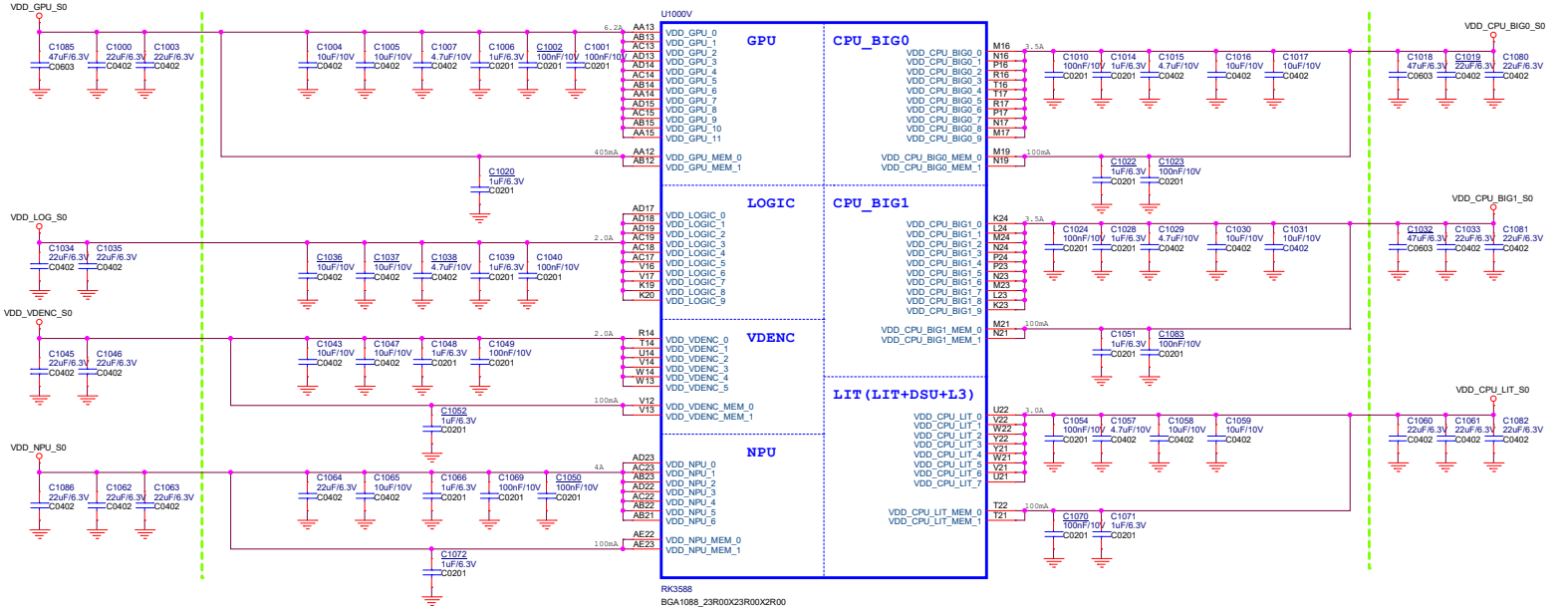
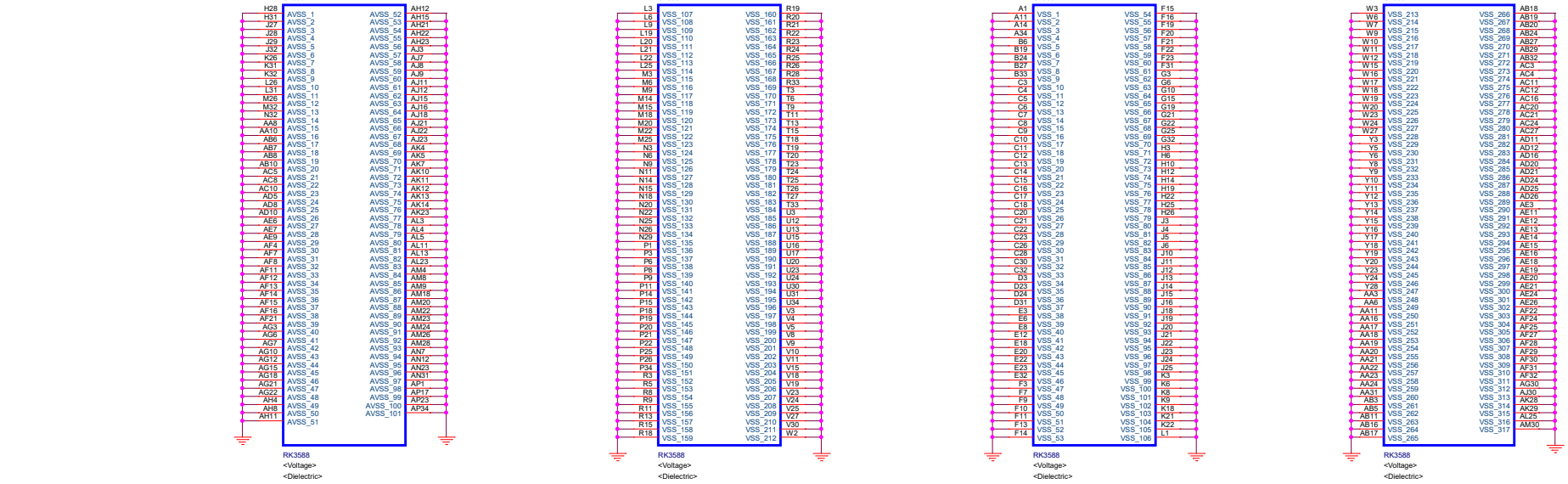
BPI-M7 Block Diagram



RK3588_V(POWER)



Note:
The Caps between green line and U1000 should be placed under the U1000 package.Other caps should be placed close to the U1000 package

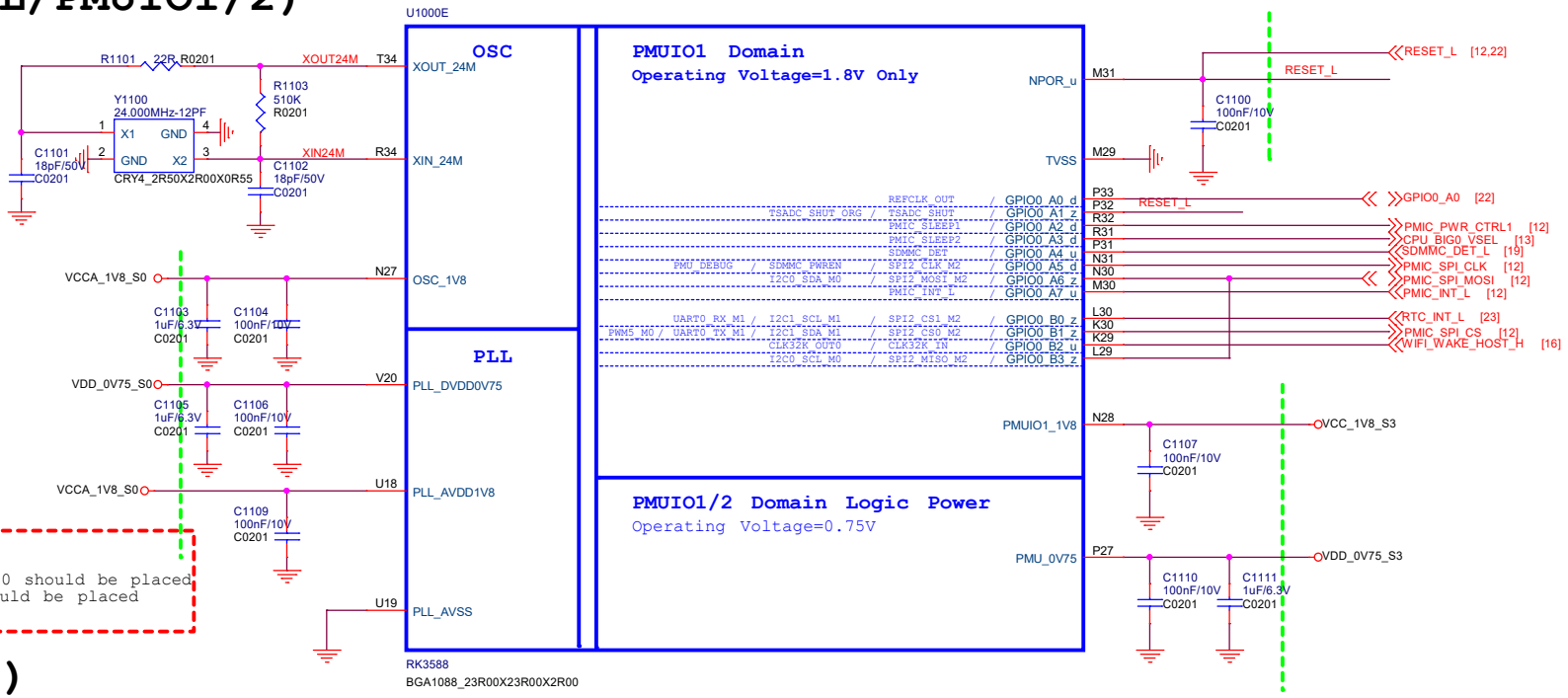


RK3588_E(OSC/PLL/PMUIO1/2)

Note:
Adjusted the load capacitance according to the crystal specification

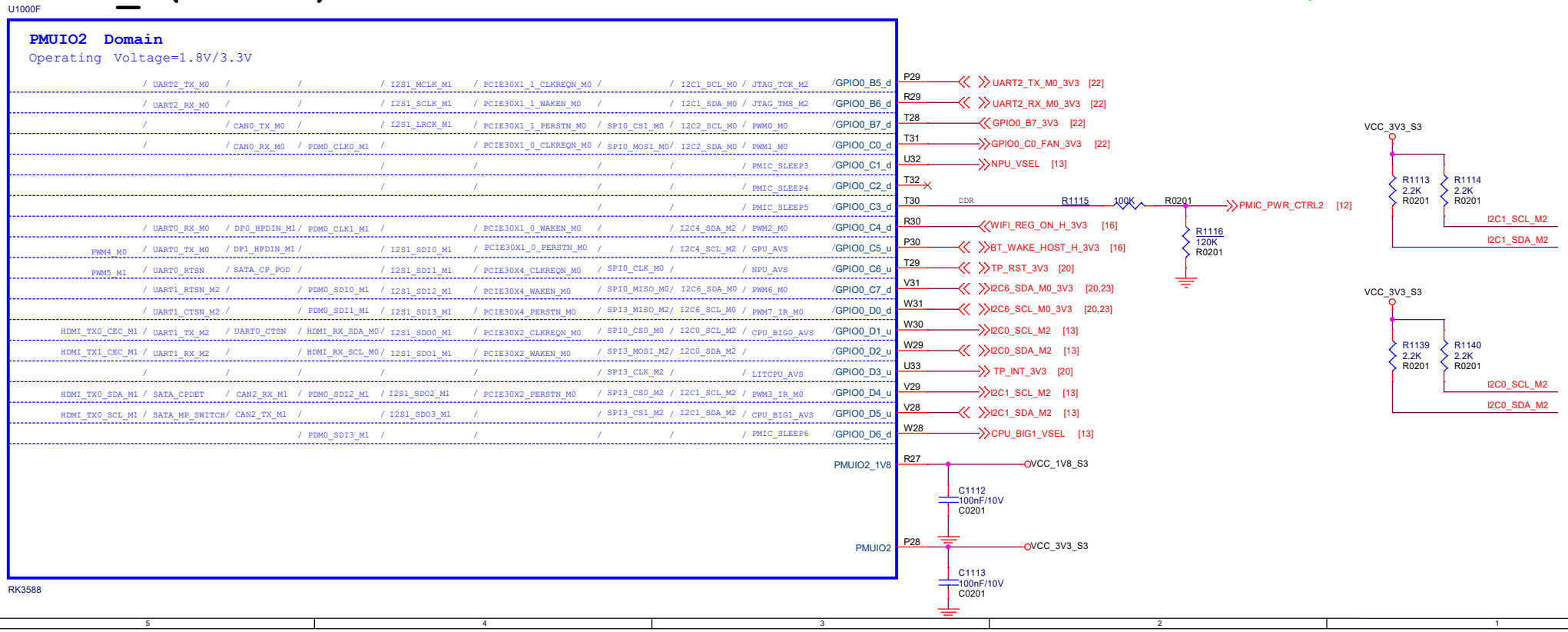
The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

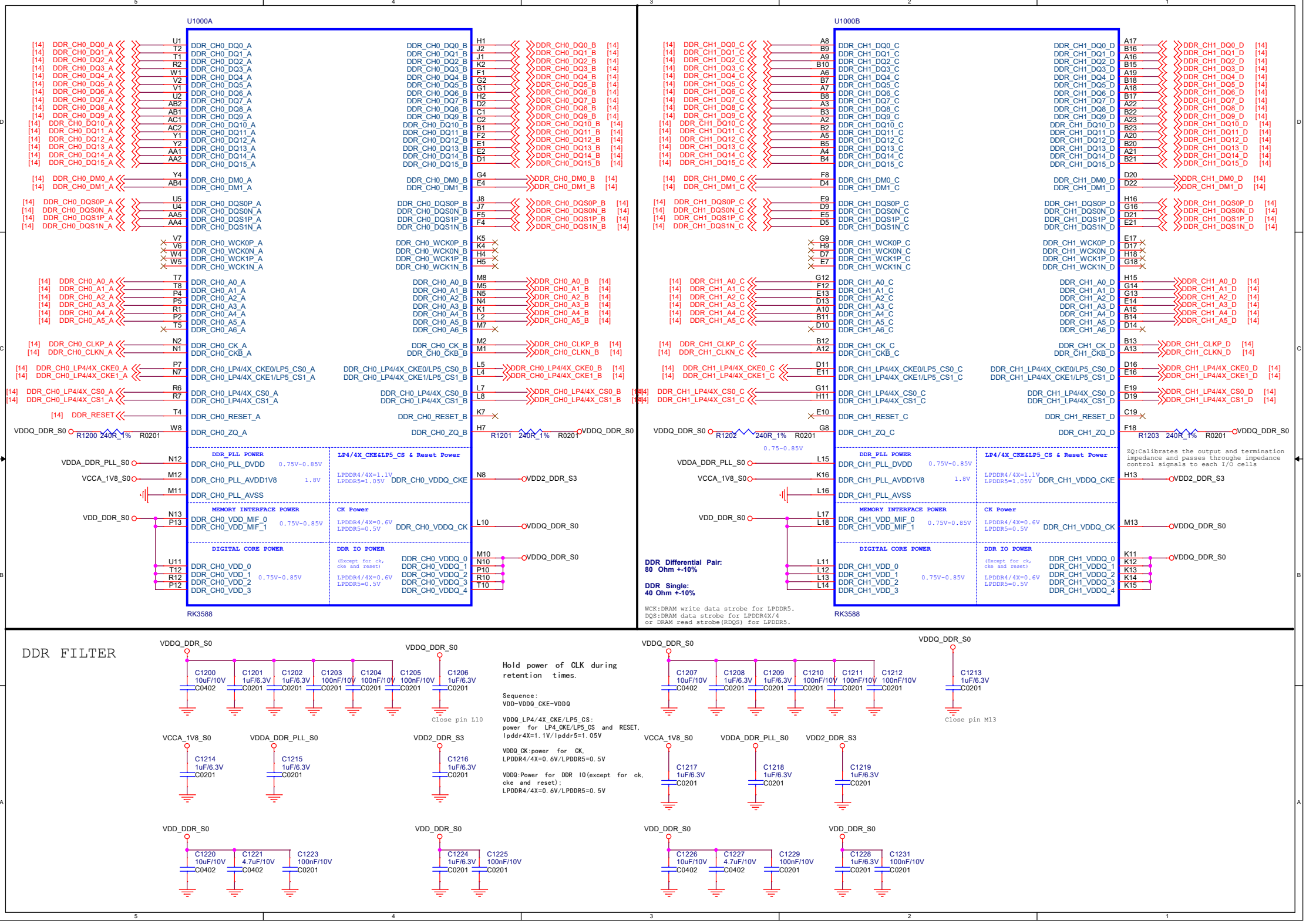
CL={CL1*CL2/(CL1+CL2))}+PCB strays
Total CL<=12pF



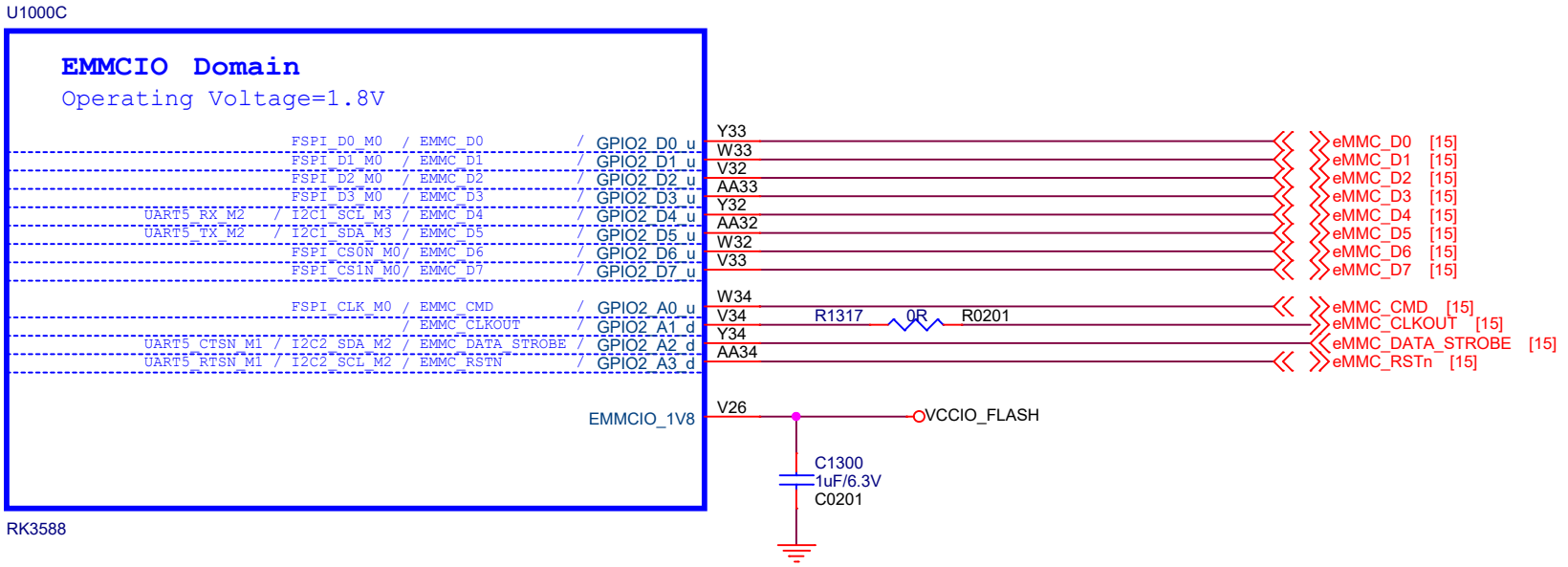
Note:
The Caps between green line and U1000 should be placed under the U1000 package.Other caps should be placed close to the U1000 package

RK3588_F(PMUIO2)

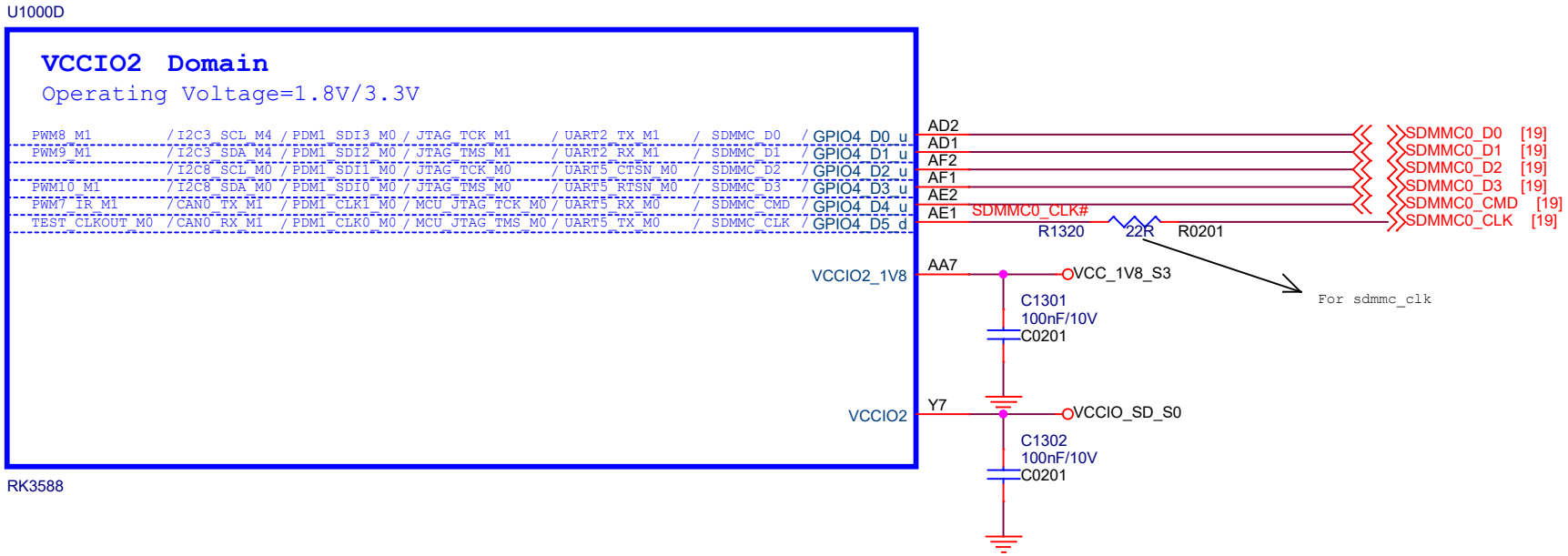



U1000A
U1000B
RK3588
DDR_PLL POWER
LP4/4X_CKE&LP5_CS & Reset Power
MEMORY INTERFACE POWER
CK Power
DIGITAL CORE POWER
DDR IO POWER
DDR Differential Pair:
80 Ohm +-10%
DDR Single:
40 Ohm +-10%
WCK:DRAM write data strobe for LPDDR5.
DQS:DRAM data strobe for LPDDR4X/4
or DRAM read strobe(RDQS) for LPDDR5.
ZQ:Calibrates the output and termination impedance and passes throughe impedance control signals to each I/O cells
DDR FILTER
Hold power of CLK during retention times.
Sequence:
VDD-VDDQ_CKE-VDDQ
VDDQ_LP4/4X_CKE/LP5_CS:
power for LP4_CKE/LP5_CS and RESET, lpddr4X=1.1V/lpddr5=1.05V
VDDQ_CK:power for CK, LPDDR4/4X=0.6V/LPDDR5=0.5V
VDDQ:Power for DDR IO(except for ck, cke and reset);
LPDDR4/4X=0.6V/LPDDR5=0.5V

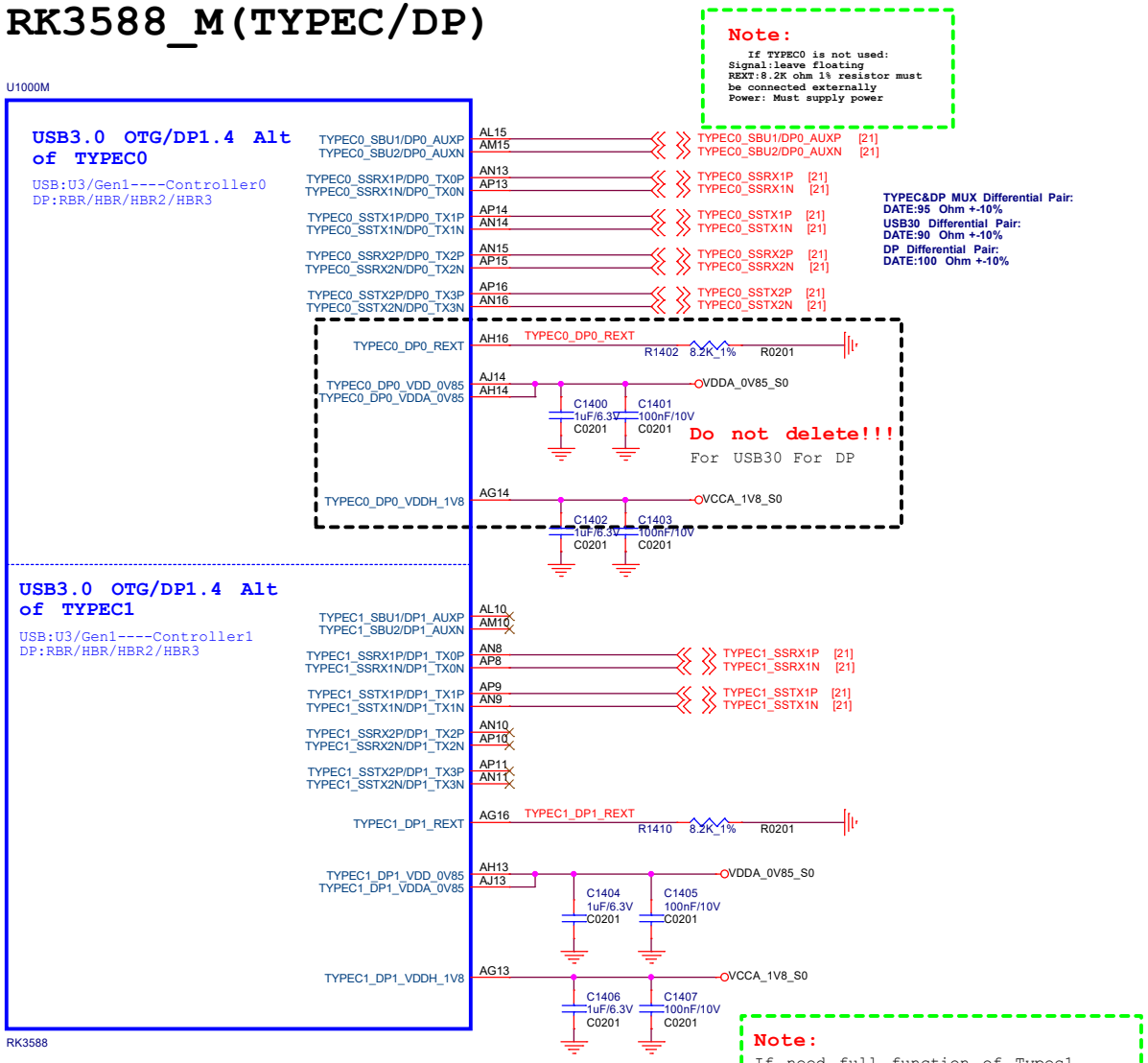
RK3588_C(EMMCIO Domain)



RK3588_D(VCCIO2 Domain)



RK3588_M(TYPEC/DP)



Note:
If TYPEC0 is not used:
Signal:leave floating
REXT:8.2K ohm 1% resistor must be connected externally
Power: Must supply power

TYPEC&DP MUX Differential Pair: DATE:95 Ohm +-10%
USB30 Differential Pair: DATE:90 Ohm +-10%
DP Differential Pair: DATE:100 Ohm +-10%

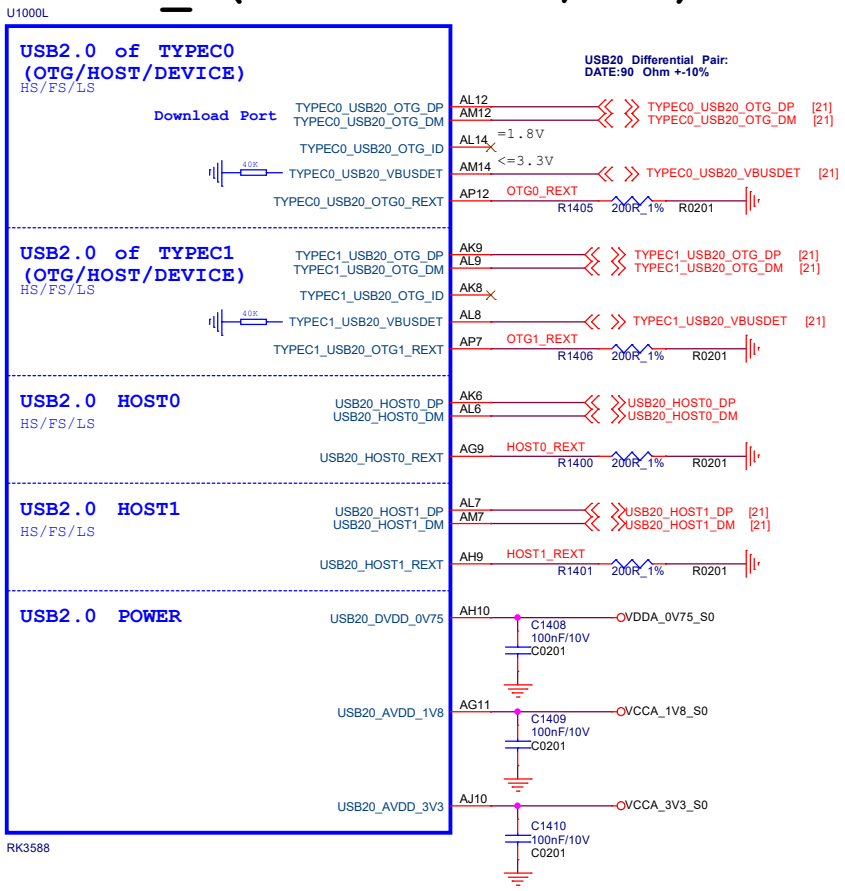
Note:
If need full function of Typec1 (With DP funtion) please Refer to the circuit of Typec0

If TYPEC1 is not used,
Signal: Leave floating
REXT: Leave floating
Power: Leave floating

USB30/DP1.4 Alt Mode Configuration

Option1	*DP x4Lane*	*DP_TX_Lane0-3*
Option2	*USB30 x4Lane*	*TYPEC_SSTX1P/1N&TYPEC_SSRX1P/1N or TYPEC_SSTX2P/2N&TYPEC_SSRX2P/2N*
Option3	*USB30X2Lane+DPX2Lane*	*USB30:Lane0 Lane1 DP:Lane2 Lane3*
Option4	*USB30X2Lane+DPX2Lane*	*USB30:Lane2 Lane3 DP:Lane0 Lane1*

RK3588_L(USB2.0 HOST/OTG)



USB20 Differential Pair: DATE:90 Ohm +-10%

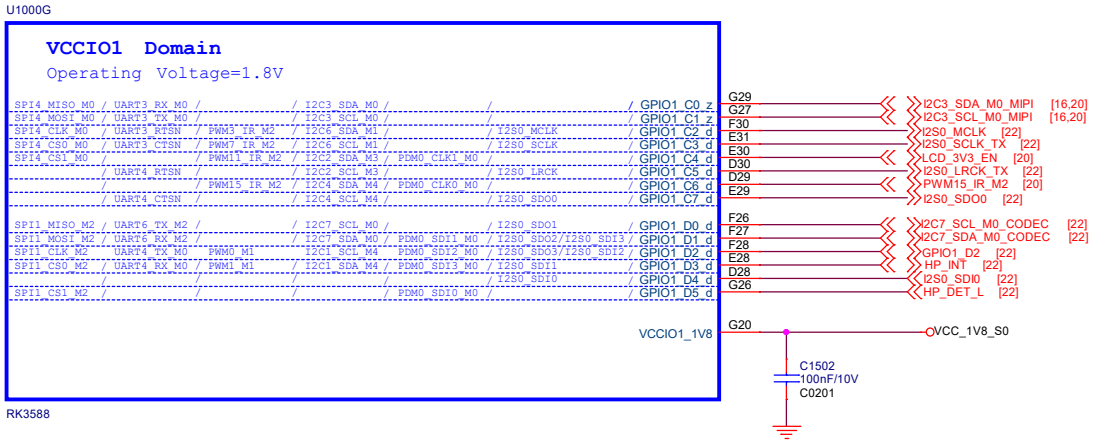
Note:
TYPEC0_USB20_OTG:
DP/DM:Must used for download
ID:According to demand,if not used,Leave floating
VBUSDET:Must provide
REXT:200ohm 1% resistor must be connected externally
Power: Must supply power

TYPEC1_USB20_OTG:
If not used:
DP/DM:Leave floating
ID:Leave floating
VBUSDET:Leave floating
REXT:Leave floating
Power:Leave floating

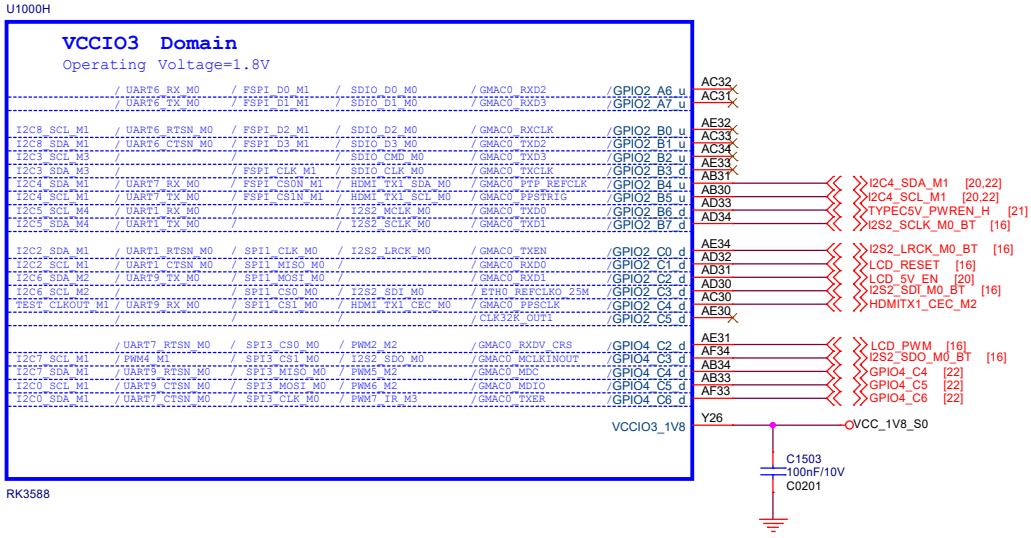
USB20_HOST0/USB20_HOST1:
If not used:
DP/DM:Leave floating
REXT:Leave floating
Power:Leave floating

Note:
The USB20_VBUSDET pin internal has a pull-down resistance(40K ohm) to ground,The resistance creates a voltage with the external series 30K ohm resistor.The VBUSDETpin voltage range <=3.3V.

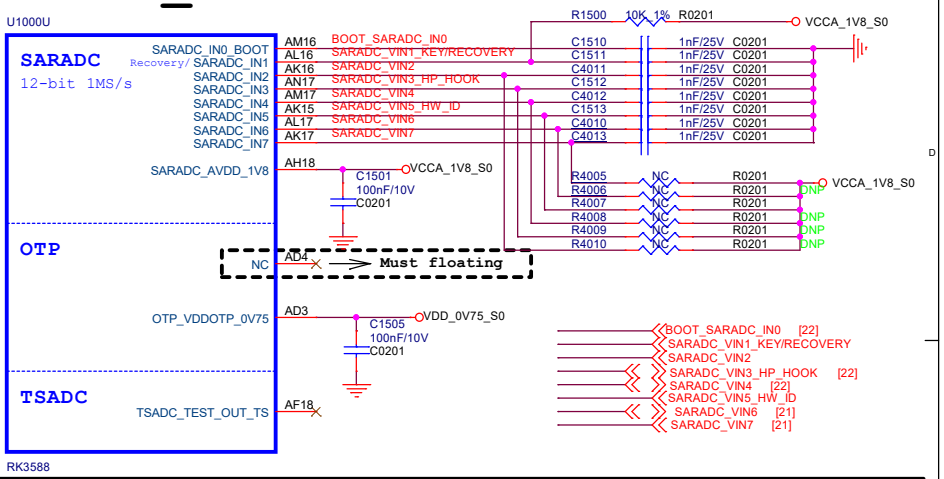
RK3588_G(VCCIO1 Domain)



RK3588_H(VCCIO3 Domain)



RK3588_U(SARADC/OTP)



BOOT MODE CONFIG

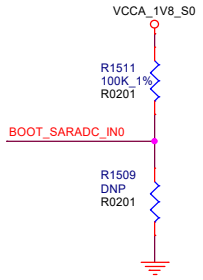


TABLE 1

Item	Rup	Rdown	ADC	VOL	BOOT MODE
LEVEL1	100K	0ohm	0	0V	USB (Maskrom mode)
LEVEL2	100K	20K	682	0.3V	SD Card-USB
LEVEL3	100K	51K	1365	0.6V	EMMC-USB
LEVEL4	100K	100K	2047	0.9V	FSPI M0-USB
LEVEL5	100K	200K	2730	1.2V	FSPI M1-USB
LEVEL6	100K	499K	3412	1.5V	FSPI M2-USB
LEVEL7	100K	DNP	4095	1.8V	FSPI_M2-FSPI_M1-FSPI_M0 -EMMC-SD Card-USB

BOARD ID CONFIG

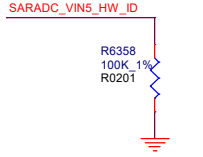
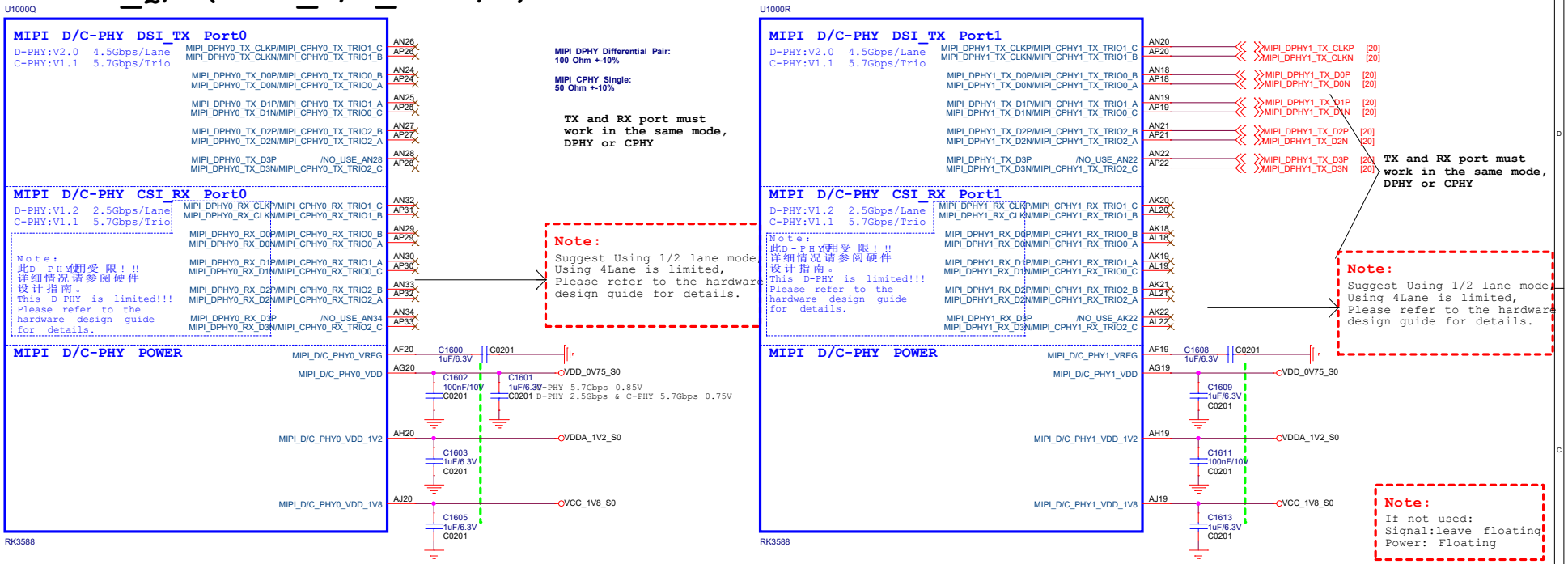


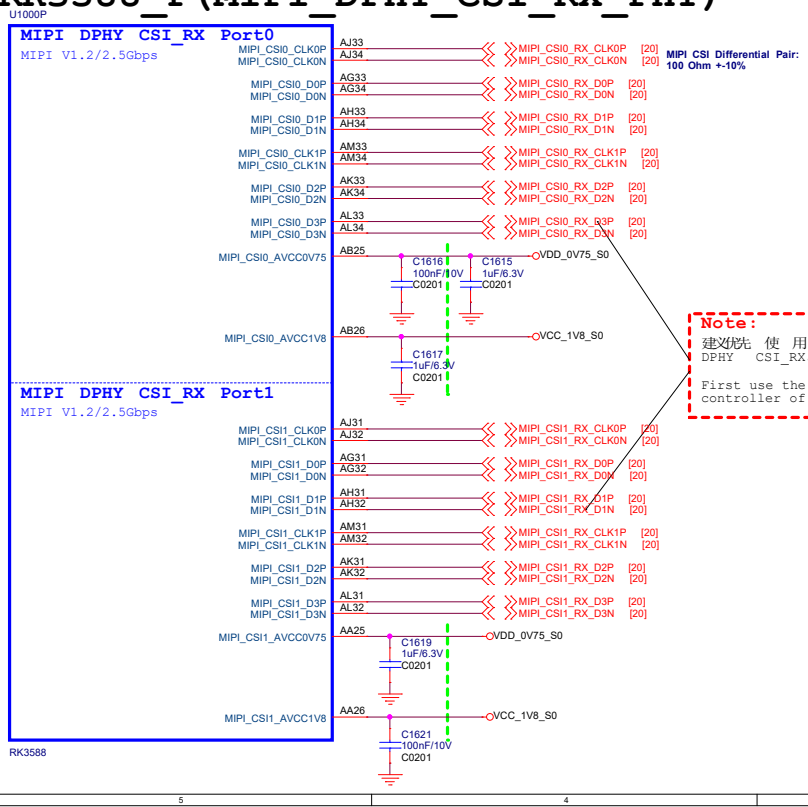
TABLE 2

Item	Rup	Rdown	ADC	VOL	VERSION
LEVEL1	DNP	100K	0	0V	V1.0
LEVEL2	100K	20K	682	0.3V	V2.0
LEVEL3	100K	51K	1365	0.6V	V3.0
LEVEL4	100K	100K	2047	0.9V	V4.0
LEVEL5	100K	200K	2730	1.2V	V5.0
LEVEL6	100K	499K	3412	1.5V	V6.0
LEVEL7	100K	DNP	4095	1.8V	V7.0

RK3588_Q/R(MIPI_D/C_PHY0/1)



RK3588_P(MIPI_DPHY_CSI_RX_PHY)



MIPI_CSI_RX Configuration

Option1	*Sensor1 x4Lane*	*MIPI_CSI_RX_D0-3* *MIPI_CSI_RX_CLK0*
Option2	*Sensor1 x2Lane* + *Sensor2 x2Lane*	*MIPI_CSI_RX_D0-1* *MIPI_CSI_RX_CLK0*
		MIPI_CSI_RX_D2-3 *MIPI_CSI_RX_CLK1*

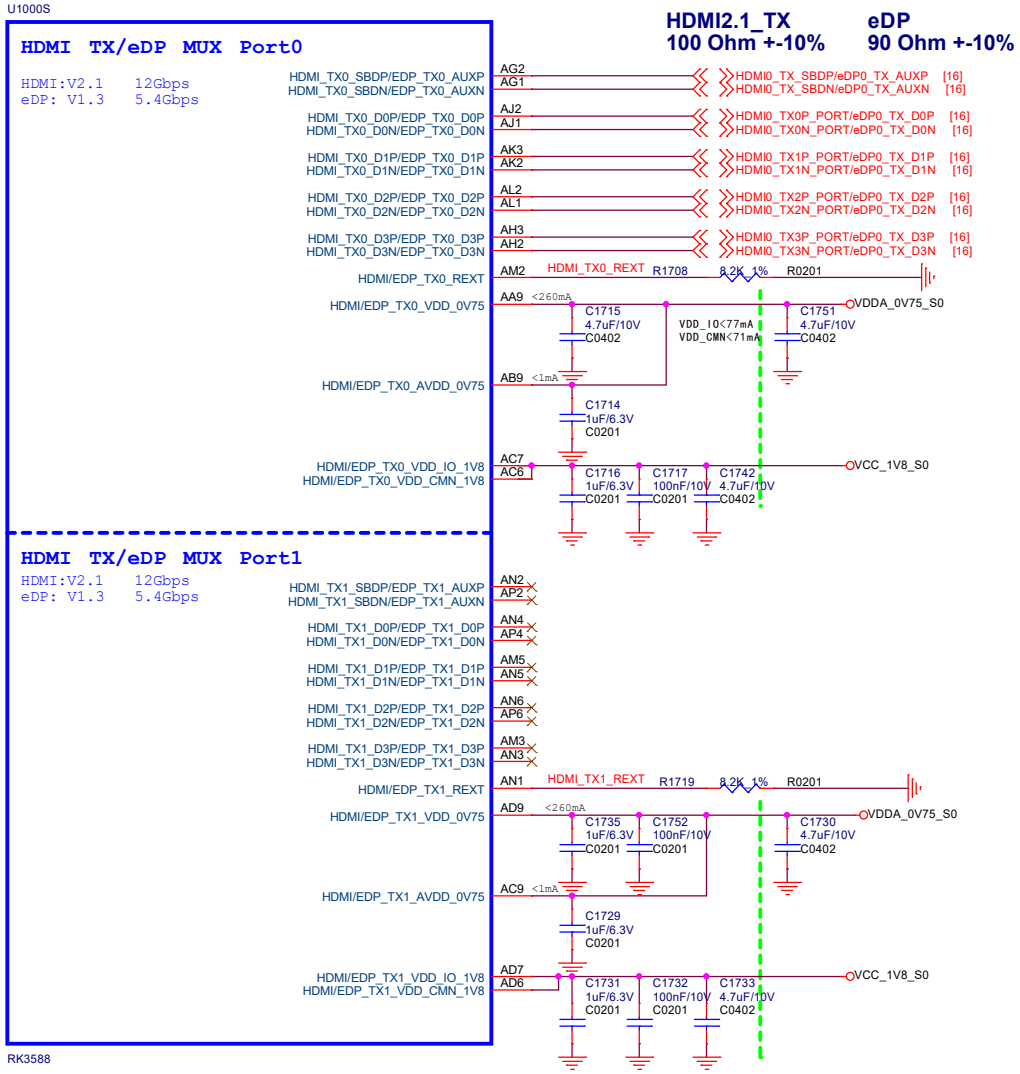
Note:
When in single clock lane mode, CLK0P/0N is the clock lane from Data lane0 to Data lane3,but clock lane1 is invalid; In dual clock lanes mode,CLK0P/0N is the clock lane of Data lane0 and Data lane1,while CLK1P/1N is the clock lane of Data lane2 and Data lane3.

Note:
The Caps to the left of green line should be placed under the U1000 package.Other caps should be placed close to the U1000 package.

Note:
If not used:
Signal:leave floating
Power: Floating

RK3588_S(HDMI2.1 TX)

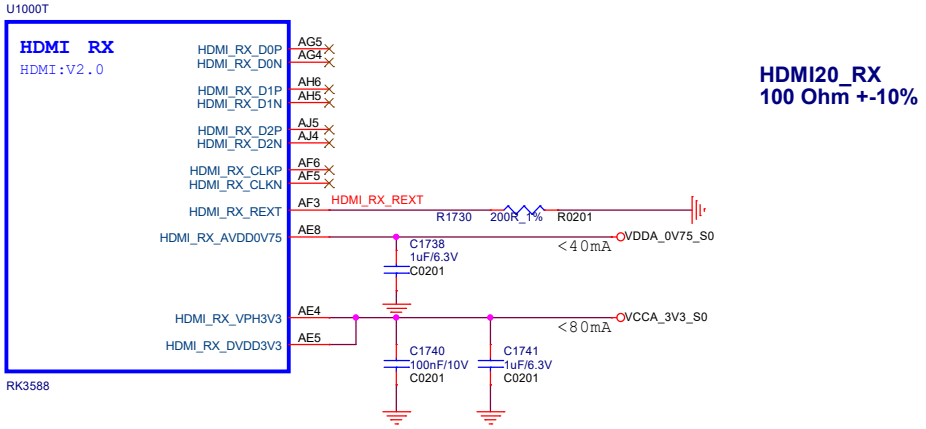
Note:
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.



Note:
The Caps to the left of green line should be placed under the U1000 package.Other caps should be placed close to the U1000 package.

Note:
If not used:
Signal:leave floating
Power: Floating or tie to VSS

RK3588_T(HDMI20 RX)



Note:
If not used:
Signal:leave floating
Power: Floating

RK3588_N(PCIE20)

U1000N

Note:
If not used:
Signal:leave floating
Power: Floating

PCIE20/SATA30 Mux0

PCIe20x1_2
1Lane(RC)--4(1L2)

SATA30 HOST
Controller0

MUX

input or output

PCIE20_0_REFCLKP / PCIE20_0_REFCLKN — L32 / L33 — PCIE20_0_REFCLKP [17] / PCIE20_0_REFCLKN [17]
PCIE20_0_TXP/SATA30_0_TXP / PCIE20_0_TXN/SATA30_0_TXN — M34 / M33 — PCIE20_0_TXP [17] / PCIE20_0_TXN [17]
PCIE20_0_RXP/SATA30_0_RXP / PCIE20_0_RXN/SATA30_0_RXN — N33 / N34 — PCIE20_0_RXP [17] / PCIE20_0_RXN [17]

PCIE20_SATA30_0_AVDD_0V85 — M28 — C1810 10uF/10V C0402 — VDDA_0V85_S0
PCIE20_SATA30_0_AVDD_1V8 — M27 — C1808 1uF/6.3V C0201 — VCCA_1V8_S0

CLK Differential Pair:
100 Ohm± 10 %
DATA Differential Pair:
PCIE20: 85 Ohm ± 10 %
SATA30: 100 Ohm ± 10 %
USB30: 90ohm ± 10 %

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30 Mux1

PCIe30x1_0
1Lane(RC)--2(1L0)

SATA30 HOST
Controller1

MUX

input or output

PCIE20_1_REFCLKP / PCIE20_1_REFCLKN — H32 / H33 — PCIE20_1_REFCLKP [18] / PCIE20_1_REFCLKN [18]
PCIE20_1_TXP/SATA30_1_TXP / PCIE20_1_TXN/SATA30_1_TXN — K33 / K34 — PCIE20_1_TXP [18] / PCIE20_1_TXN [18]
PCIE20_1_RXP/SATA30_1_RXP / PCIE20_1_RXN/SATA30_1_RXN — J33 / J34 — PCIE20_1_RXP [18] / PCIE20_1_RXN [18]

PCIE20_SATA30_1_AVDD_0V85 — L28 — C1818 100nF/10V C0201 — VDDA_0V85_S0
PCIE20_SATA30_1_AVDD_1V8 — L27 — C1816 100nF/10V C0201 — VCCA_1V8_S0

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30/USB30 HOST Mux2

PCIe30x1_1
1Lane(RC)--3(1L1)

SATA30 HOST
Controller2

USB30 HOST
Controller2

MUX

input or output

PCIE20_2_REFCLKP / PCIE20_2_REFCLKN — G31 / G30 — PCIE20_2_REFCLKP [16] / PCIE20_2_REFCLKN [16]
PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP / PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN — H30 / H29 — PCIE20_2_TXP [16] / PCIE20_2_TXN [16]
PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP / PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN — J31 / J30 — PCIE20_2_RXP [16] / PCIE20_2_RXN [16]

PCIE20_SATA30_USB30_2_AVDD_0V85 — K28 — C1806 1uF/6.3V C0201 — VDDA_0V85_S0
PCIE20_SATA30_USB30_2_AVDD_1V8 — K27 — C1804 1uF/6.3V C0201 — VCCA_1V8_S0

PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3

RK3588

Note:
The SATA differential trace impedance is 100 OHM
The SATA trace length is less than 5 inch

RK3588_O(PCIE30)

Note:
Only PCIE3.0 Controller 0 support RC and EP,Other controller only support RC Mode.

U1000O

PCIE30X4

PCIe30X4
4Lane(DM)--0(4L)

PCIe30x1_0
1Lane(RC)--2(1L0)

PCIE20/SATA30 MUX1

PCIe30x2
2Lane(RC)--1(2L)

PCIe30x1_1
1Lane(RC)--3(1L1)

PCIE20/SATA30/USB30 MUX2

PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)

PCIE30 PORT0

input PCIE30_PORT0_REF_CLKP / PCIE30_PORT0_REF_CLKN — E33 / E34 — PCIE30_PORT0_REFCLKP_IN [19] / PCIE30_PORT0_REFCLKN_IN [19]

Lane0
PCIE30_PORT0_TX0P / PCIE30_PORT0_TX0N — D32 / D33 — PCIE30_PORT0_TX0P [19] / PCIE30_PORT0_TX0N [19]
PCIE30_PORT0_RX0P / PCIE30_PORT0_RX0N — G33 / G34 — PCIE30_PORT0_RX0P [19] / PCIE30_PORT0_RX0N [19]

Lane1
PCIE30_PORT0_TX1P / PCIE30_PORT0_TX1N — C33 / C34 — PCIE30_PORT0_TX1P [19] / PCIE30_PORT0_TX1N [19]
PCIE30_PORT0_RX1P / PCIE30_PORT0_RX1N — F32 / F33 — PCIE30_PORT0_RX1P [19] / PCIE30_PORT0_RX1N [19]

PCIE30_PORT0_RESREF — B34 — PCIE30_0_REF — R1801 200R 1% R0201
PCIE30_PORT0_AVDD0V75 — G24 — VDD_0V75_S0
C1802 1uF/6.3V C0201 / C1803 4.7uF/10V C0402
PCIE30_PORT0_AVDD1V8 — G23 — VCCA_1V8_S0
C1800 1uF/6.3V C0201 / C1801 4.7uF/10V C0402

PCIE30 PORT1

input PCIE30_PORT1_REF_CLKP / PCIE30_PORT1_REF_CLKN — A28 / B28 — PCIE30_PORT1_REFCLKP_IN [19] / PCIE30_PORT1_REFCLKN_IN [19]

Lane0
PCIE30_PORT1_TX0P / PCIE30_PORT1_TX0N — B30 / A30 — PCIE30_PORT1_TX2P [19] / PCIE30_PORT1_TX2N [19]
PCIE30_PORT1_RX0P / PCIE30_PORT1_RX0N — B32 / A32 — PCIE30_PORT1_RX2P [19] / PCIE30_PORT1_RX2N [19]

Lane1
PCIE30_PORT1_TX1P / PCIE30_PORT1_TX1N — C29 / B29 — PCIE30_PORT1_TX3P [19] / PCIE30_PORT1_TX3N [19]
PCIE30_PORT1_RX1P / PCIE30_PORT1_RX1N — C31 / B31 — PCIE30_PORT1_RX3P [19] / PCIE30_PORT1_RX3N [19]

PCIE30_PORT1_RESREF — A33 — PCIE30_1_REF — R1807 200R 1% R0201
PCIE30_PORT1_AVDD0V75 — H24 — VDD_0V75_S0
C1814 1uF/6.3V C0201 / C1815 4.7uF/10V C0402
PCIE30_PORT1_AVDD1V8 — H23 — VCCA_1V8_S0
C1812 100nF/10V C0201 / C1813 4.7uF/10V C0402

CLK Differential Pair:
100 Ohm ± 10 %
DATA Differential Pair:
PCIE30: 85 Ohm ± 10 %

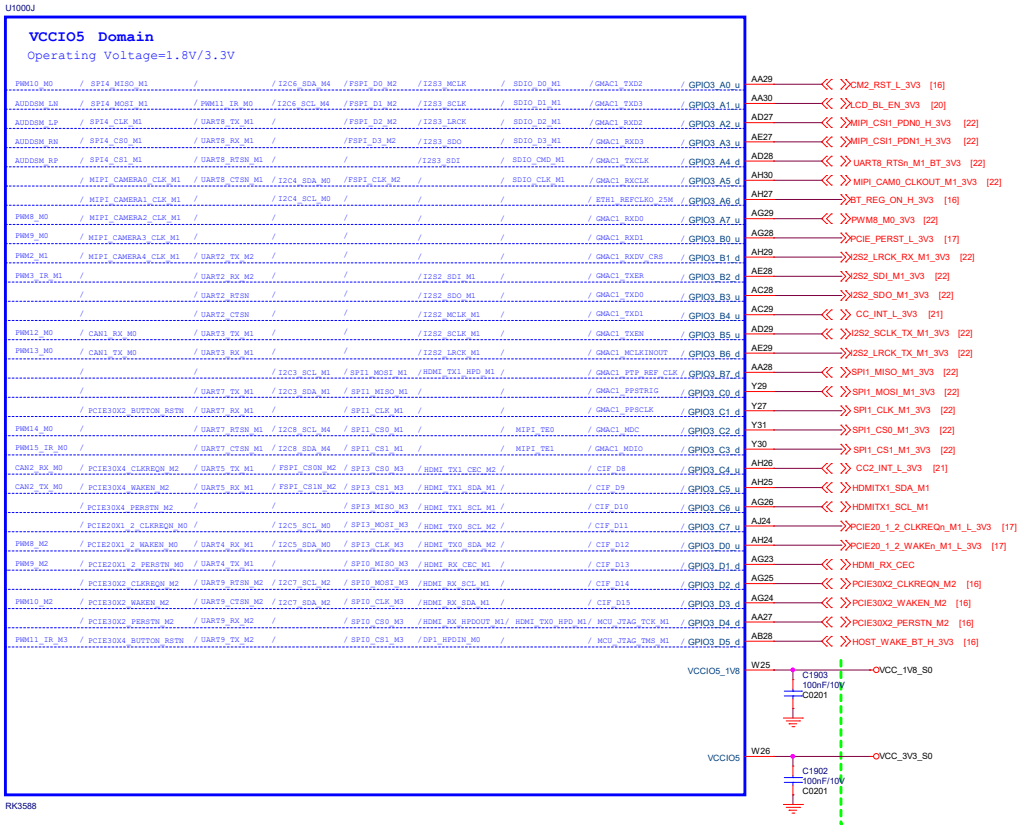
RK3588

Note:
If Port0 and Port1 are not used,
Port0 and Port1 REF_CLKP/N: Leave floating or tie to VSS
Port0 and Port1 Other Signal: Leave floating
Port0 and Port1 Power: Leave floating or tie to VSS

If Port0 is used ,Port1 is not used,
Port1 REF_CLKP/N: Leave floating or tie to VSS
Port1 Other Signal: Leave floating
Port1 Power: Must supply power

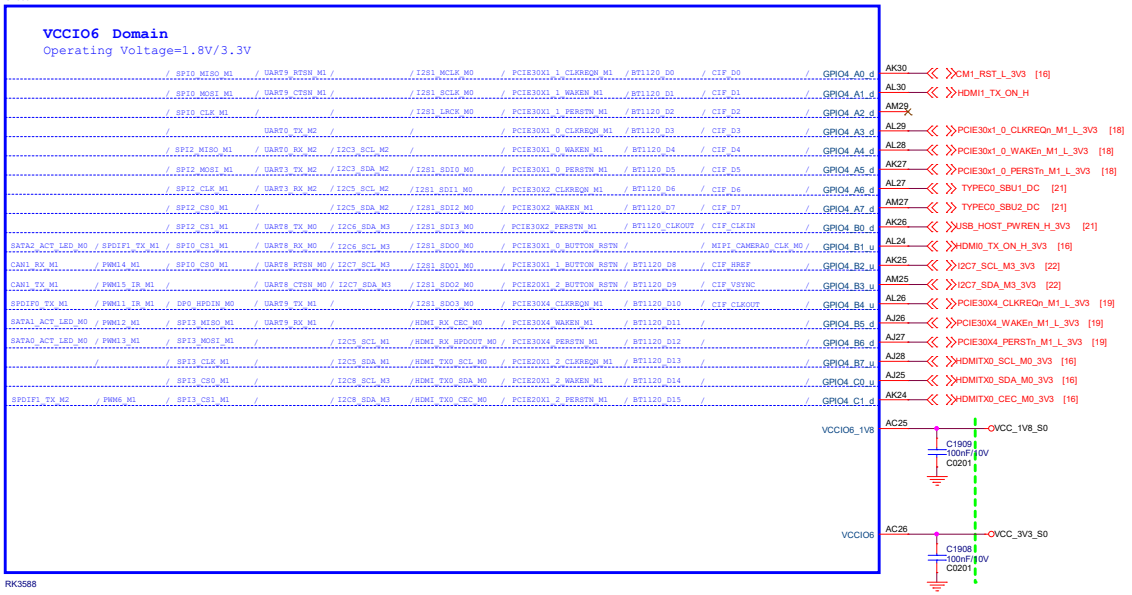
If Port1 is used ,Port0 is not used,
Port0 REF_CLKP/N: Leave floating or tie to VSS
Port0 Other Signal: Leave floating
Port0 Power: Must supply power

RK3588_J(VCCIO5 Domain)



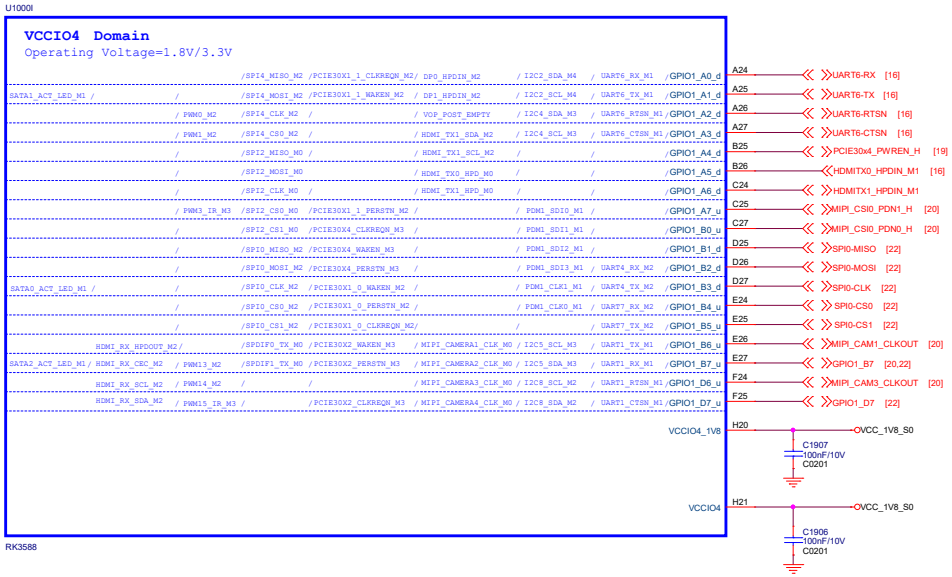
RK3588

RK3588_K(VCCIO6 Domain)



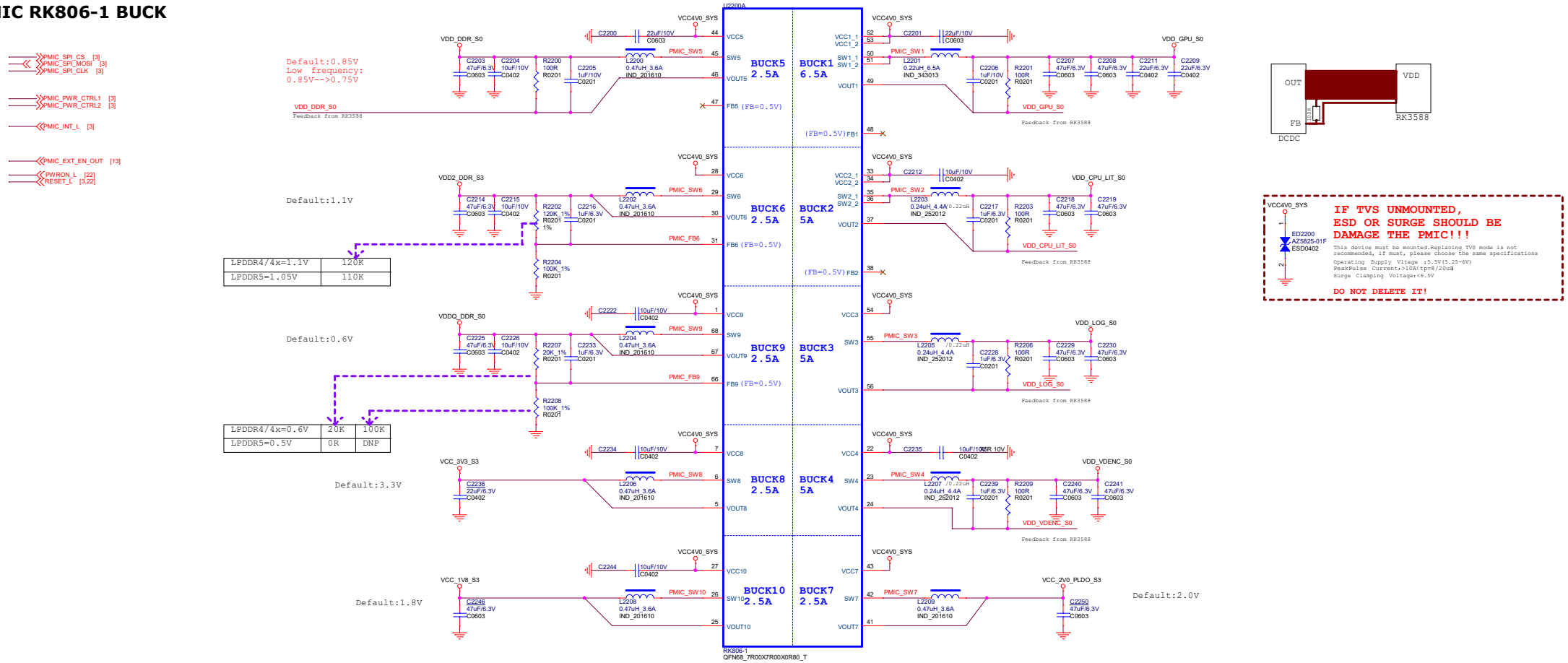
RK3588

RK3588_I(VCCIO4 Domain)

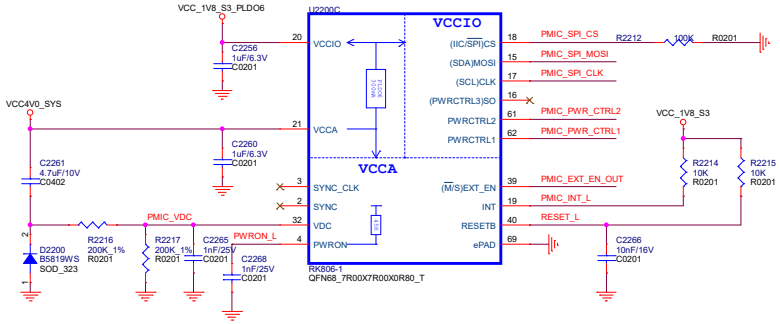


RK3588

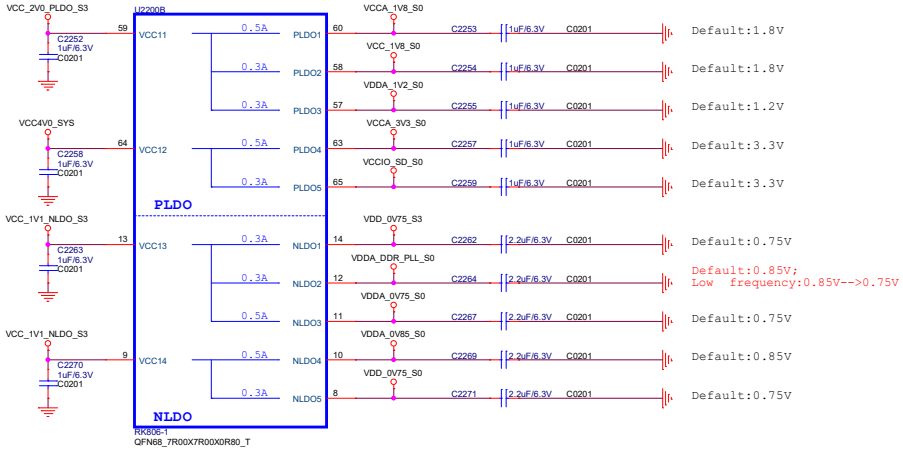
PMIC RK806-1 BUCK



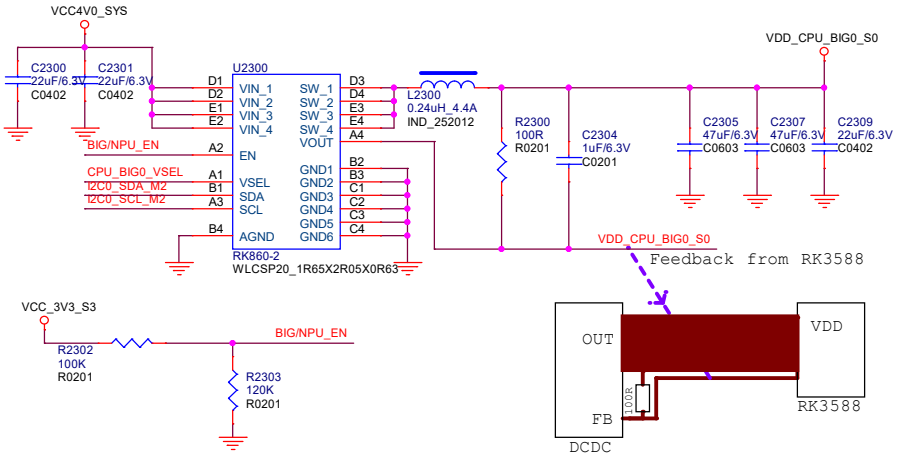
PMIC RK806-1 Managerment



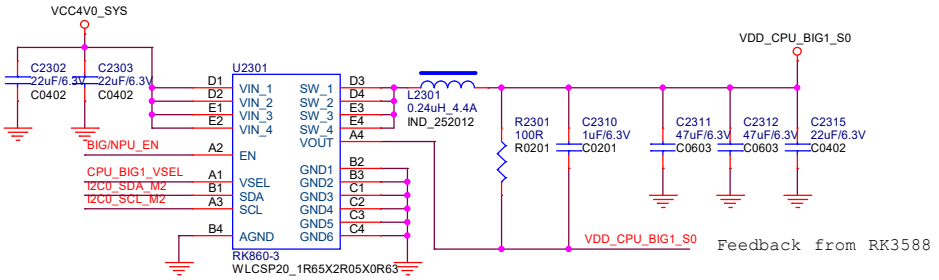
PMIC RK806-1 LDO



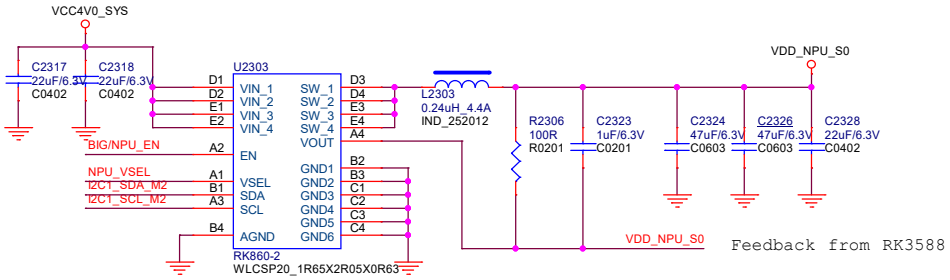
VDD_CPU_BIG0



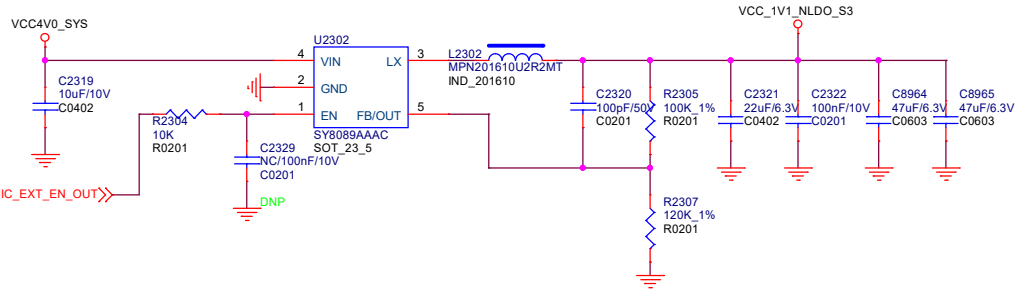
VDD_CPU_BIG1

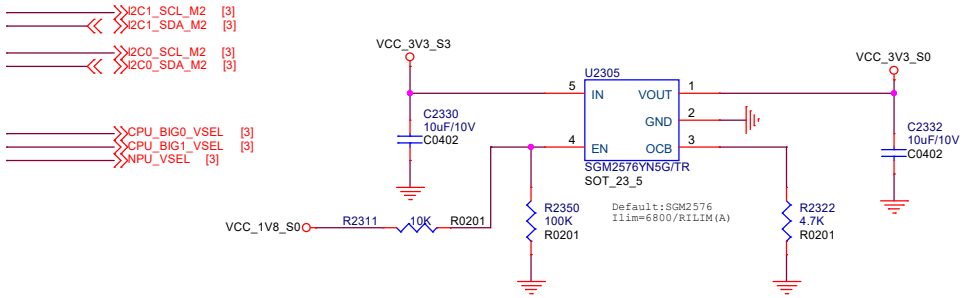


VDD_NPU

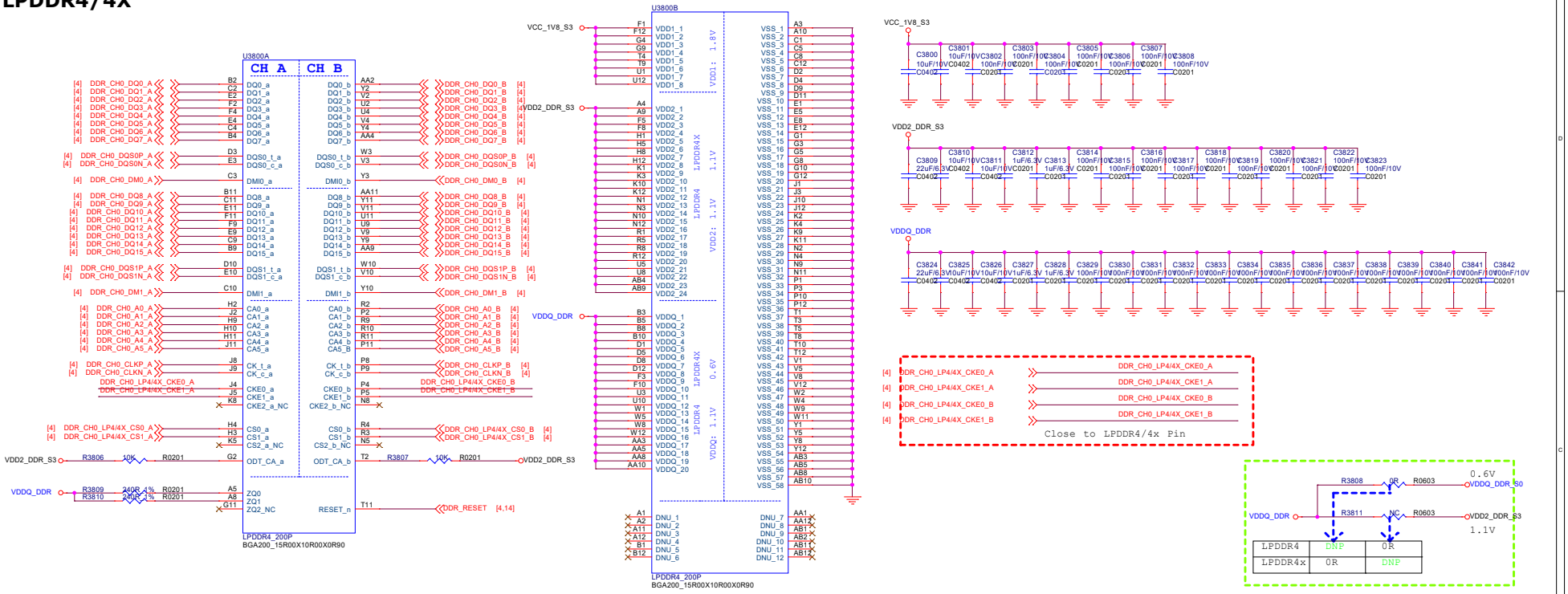


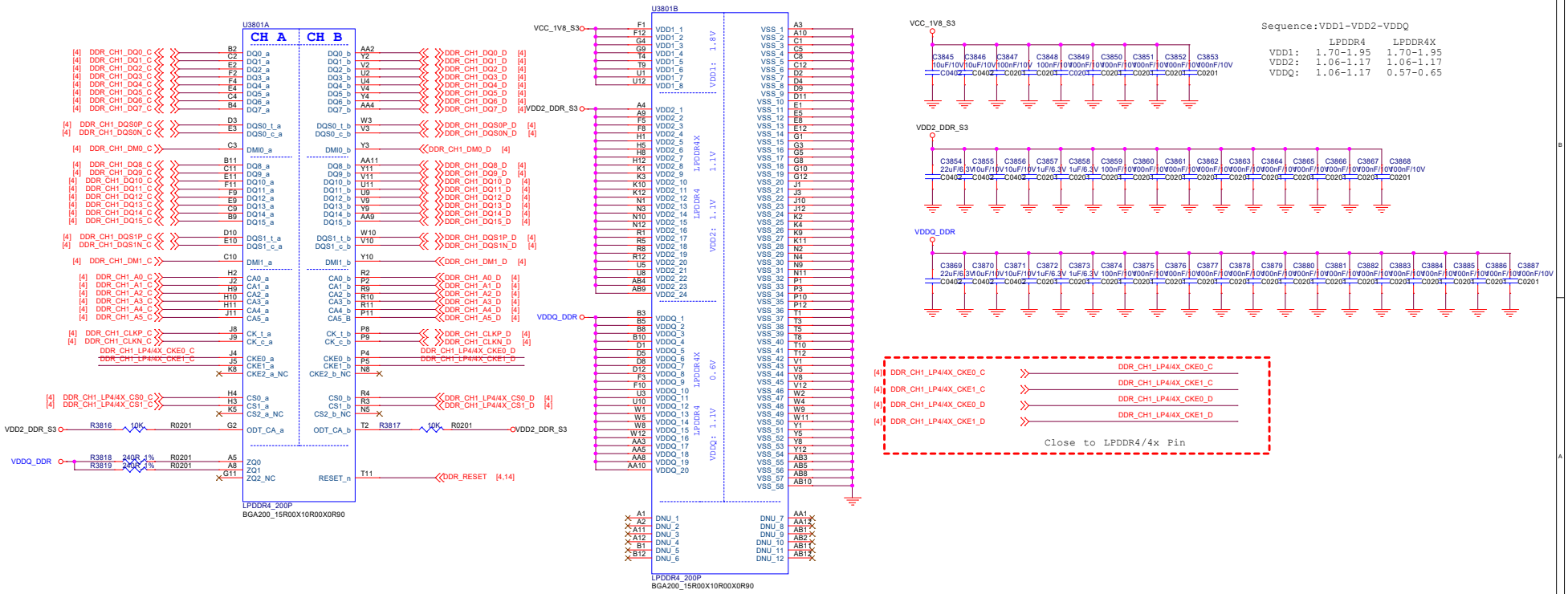
VCC_1V1_NLDO_S3



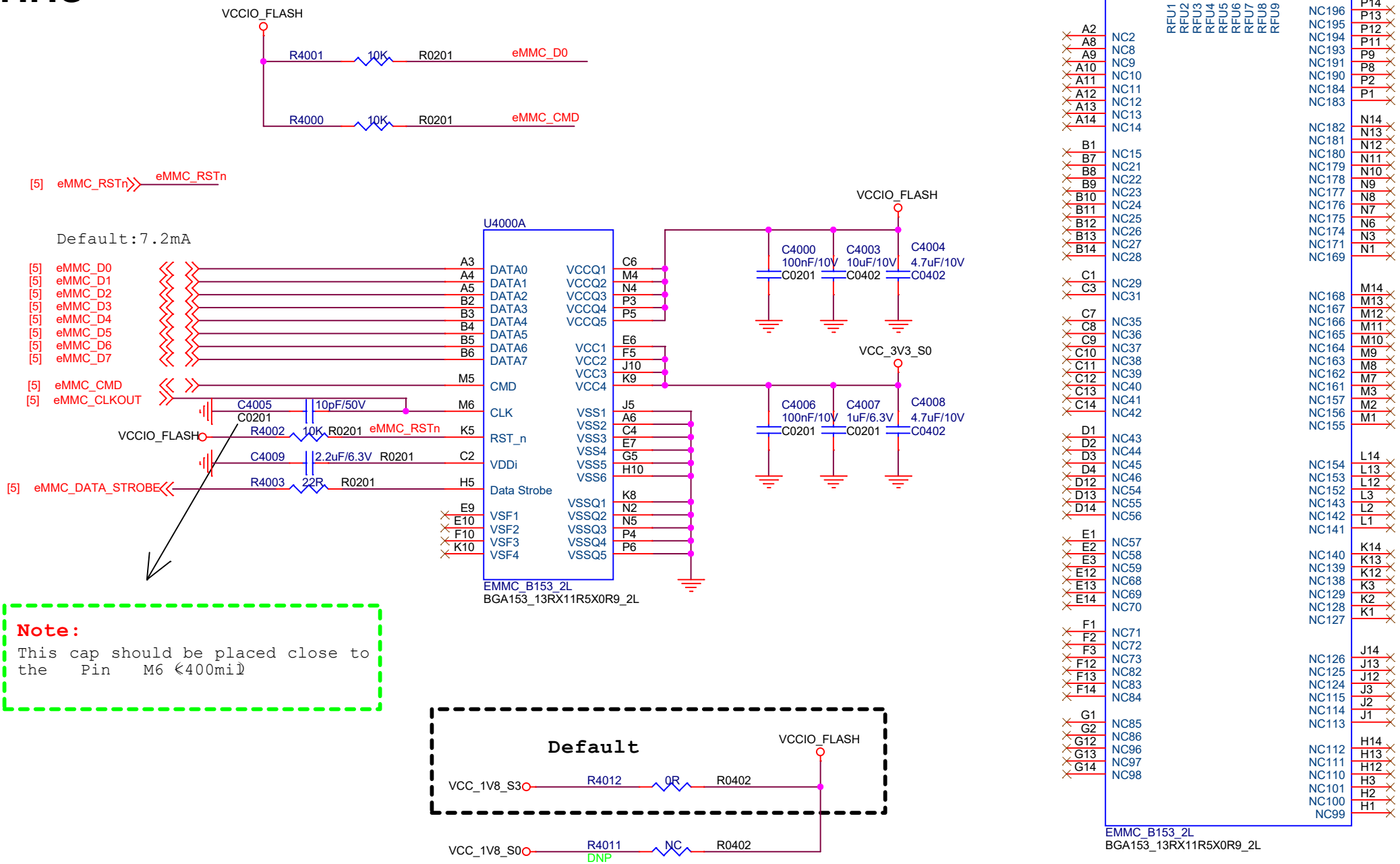


LPDDR4/4X

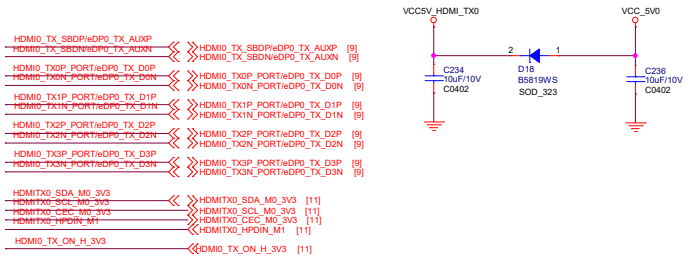

U3800A
CH A
CH B
U3800B
LPDDR4_200P
BGA200_15R00X10R00X0R90
VCC_1V8_S3
VDD2_DDR_S3
VDDQ_DDR
VDD1: 1.8V
VDD2: 1.1V
VDDQ: 0.6V
LPDDR4
LPDDR4X
Close to LPDDR4/4x Pin
0.6V
1.1V
LPDDR4 DNP 0R
LPDDR4x 0R DNP


U3801A
CH A
CH B
U3801B
LPDDR4_200P
BGA200_15R00X10R00X0R90
VCC_1V8_S3
VDD2_DDR_S3
VDDQ_DDR
VDD1: 1.8V
VDD2: 1.1V
VDDQ: 0.6V
LPDDR4
LPDDR4X
Close to LPDDR4/4x Pin
Sequence:VDD1-VDD2-VDDQ
LPDDR4 LPDDR4X
VDD1: 1.70-1.95 1.70-1.95
VDD2: 1.06-1.17 1.06-1.17
VDDQ: 1.06-1.17 0.57-0.65

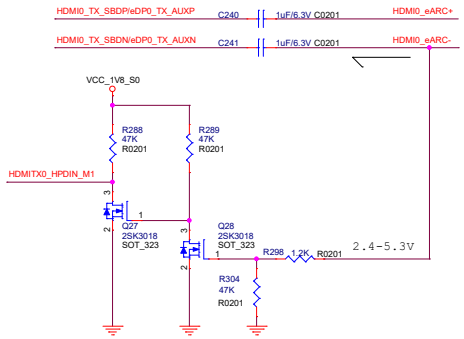
eMMC


VCCIO_FLASH
R4001 10K R0201 eMMC_D0
R4000 10K R0201 eMMC_CMD
[5] eMMC_RSTn eMMC_RSTn
Default:7.2mA
[5] eMMC_D0
[5] eMMC_D1
[5] eMMC_D2
[5] eMMC_D3
[5] eMMC_D4
[5] eMMC_D5
[5] eMMC_D6
[5] eMMC_D7
[5] eMMC_CMD
[5] eMMC_CLKOUT
[5] eMMC_DATA_STROBE
C4005 10pF/50V C0201
R4002 10K R0201 eMMC_RSTn
C4009 2.2uF/6.3V R0201
R4003 22R R0201
U4000A
DATA0 DATA1 DATA2 DATA3 DATA4 DATA5 DATA6 DATA7
CMD
CLK
RST_n
VDDi
Data Strobe
VSF1 VSF2 VSF3 VSF4
VCCQ1 VCCQ2 VCCQ3 VCCQ4 VCCQ5
VCC1 VCC2 VCC3 VCC4
VSS1 VSS2 VSS3 VSS4 VSS5 VSS6
VSSQ1 VSSQ2 VSSQ3 VSSQ4 VSSQ5
EMMC_B153_2L
BGA153_13RX11R5X0R9_2L
VCCIO_FLASH
C4000 100nF/10V C0201
C4003 10uF/10V C0402
C4004 4.7uF/10V C0402
VCC_3V3_S0
C4006 100nF/10V C0201
C4007 1uF/6.3V C0201
C4008 4.7uF/10V C0402
Note:
This cap should be placed close to the Pin M6 (400mil)
Default
VCCIO_FLASH
VCC_1V8_S3 R4012 0R R0402
VCC_1V8_S0 R4011 NC R0402
DNP
U4000B
RFU1 RFU2 RFU3 RFU4 RFU5 RFU6 RFU7 RFU8 RFU9
NC2 NC8 NC9 NC10 NC11 NC12 NC13 NC14
NC15 NC21 NC22 NC23 NC24 NC25 NC26 NC27 NC28
NC29 NC31
NC35 NC36 NC37 NC38 NC39 NC40 NC41 NC42
NC43 NC44 NC45 NC46 NC54 NC55 NC56
NC57 NC58 NC59 NC68 NC69 NC70
NC71 NC72 NC73 NC82 NC83 NC84
NC85 NC86 NC96 NC97 NC98
NC196 NC195 NC194 NC193 NC191 NC190 NC184 NC183
NC182 NC181 NC180 NC179 NC178 NC177 NC176 NC175 NC174 NC171 NC169
NC168 NC167 NC166 NC165 NC164 NC163 NC162 NC161 NC157 NC156 NC155
NC154 NC153 NC152 NC143 NC142 NC141
NC140 NC139 NC138 NC129 NC128 NC127
NC126 NC125 NC124 NC115 NC114 NC113
NC112 NC111 NC110 NC101 NC100 NC99
EMMC_B153_2L
BGA153_13RX11R5X0R9_2L

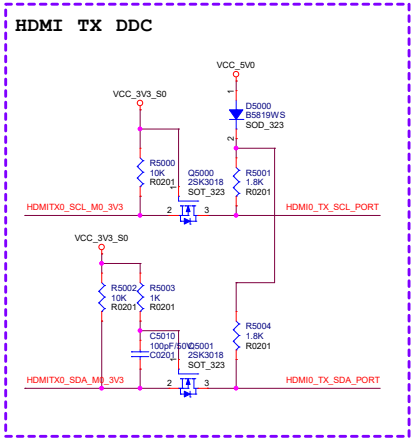
HDMI TX0



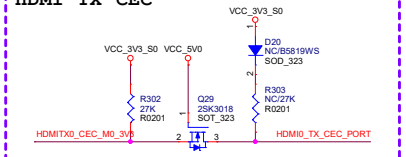
HDMI TX eARC



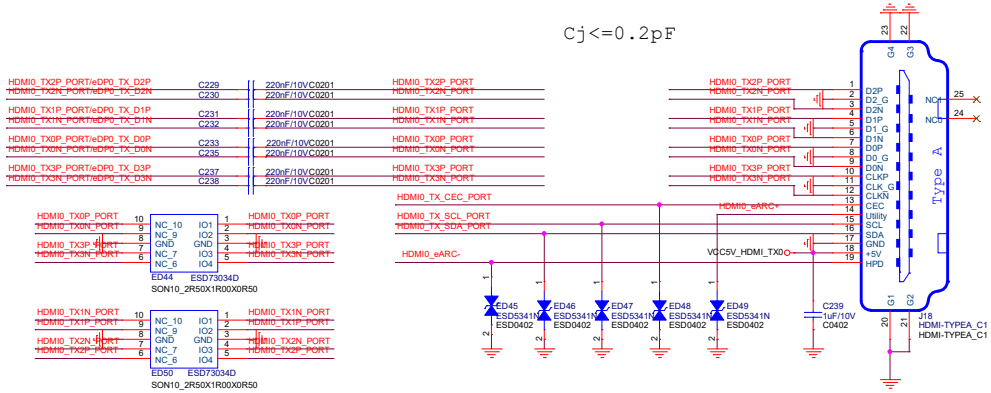
HDMI TX DDC

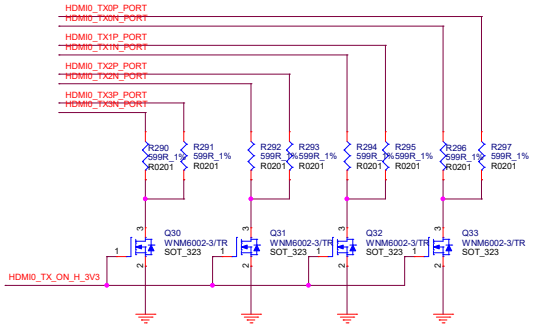


HDMI TX CEC



Cj<=0.2pF





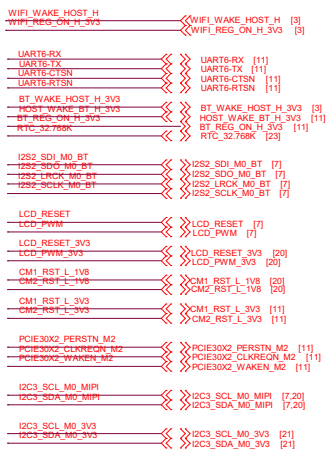
Note:

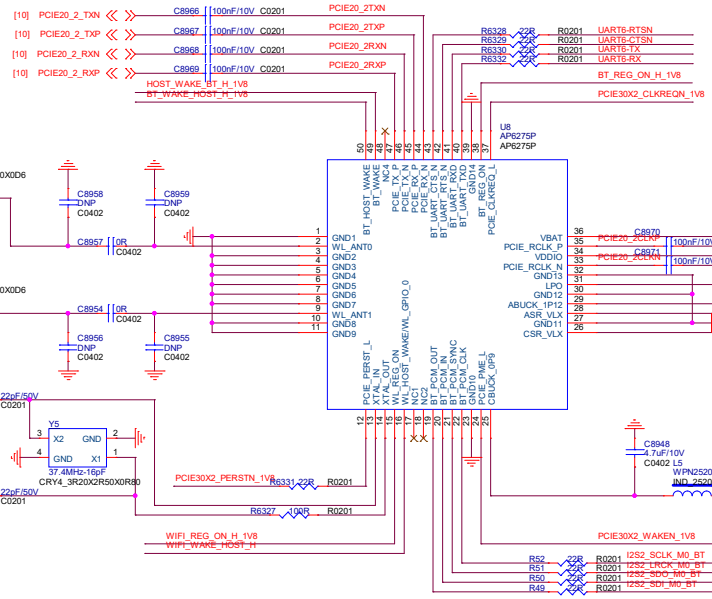
The controller only support AC coupled link In order to backward compatibility or to meet HDMI2.0 (1.4b) DC common mode spec and Voff, need do R based level-shift.

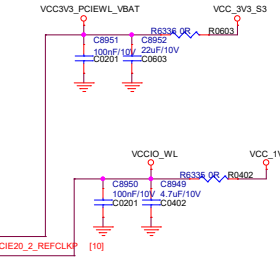
Switch on in HDMI2.0(TMDS) mode
Switch off in HDMI2.1(FRL) mode.

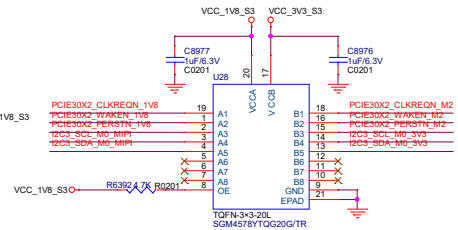
PCIE WIFI6/BT Module-2T2R

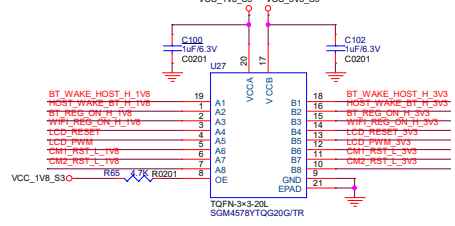
VBAT:(3.1-3.8V)/1.2A
VDDIO:(1.68-1.98V)/300mA.



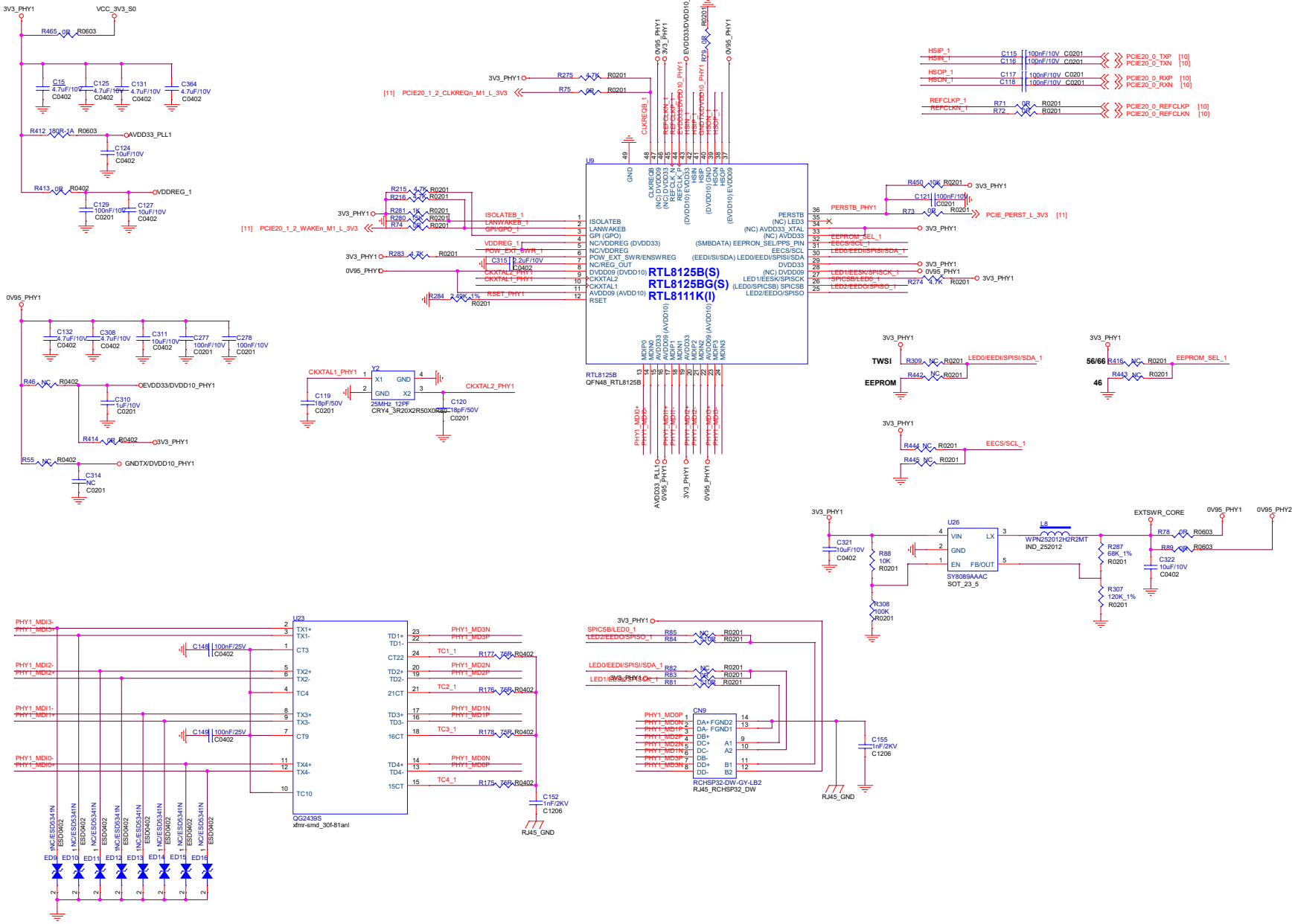




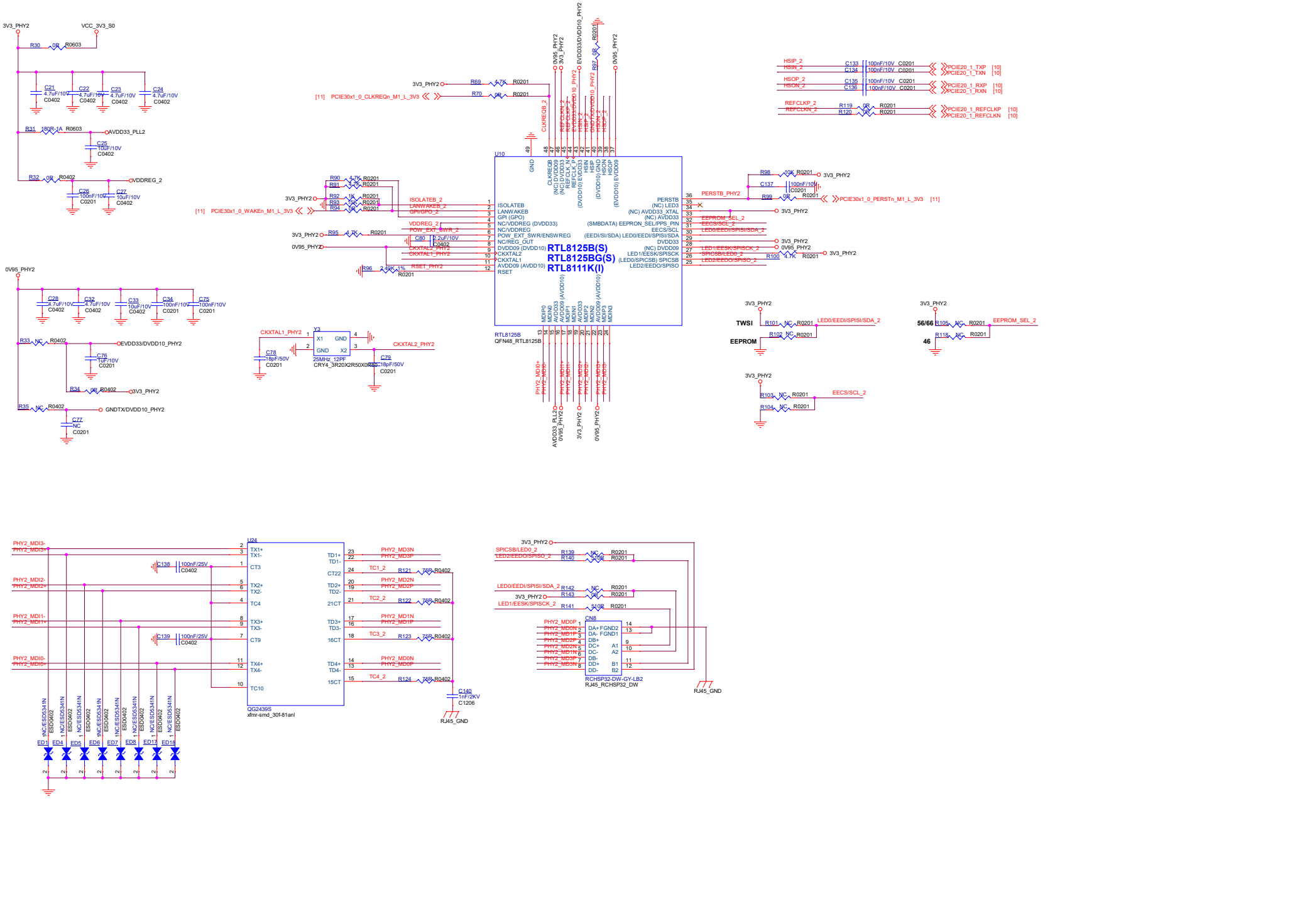




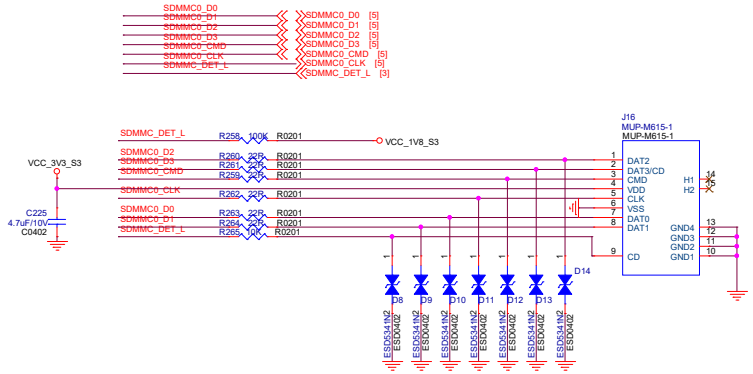
PCIE TO RJ45


3V3_PHY1
VCC_3V3_S0
R465 0R R0603
C15 4.7uF/10V C0402
C125 4.7uF/10V C0402
C131 4.7uF/10V C0402
C364 4.7uF/10V C0402
R412 180R_1A R0603
AVDD33_PLL1
C124 10uF/10V C0402
R413 0R R0402
VDDREG_1
C129 100nF/10V C0201
C127 10uF/10V C0402
0V95_PHY1
C132 4.7uF/10V C0402
C308 4.7uF/10V C0402
C311 10uF/10V C0402
C277 100nF/10V C0201
C278 100nF/10V C0201
R46 NC R0402
EVDD33/DVDD10_PHY1
C310 1uF/10V C0201
R414 0R R0402
R55 NC R0402
GNDTX/DVDD10_PHY1
C314 NC C0201
R275 4.7K R0201
[11] PCIE20_1_2_CLKREQn_M1_L_3V3
R75 0R R0201
R79 0R R0201
CLKREQB_1
REFCLK_N_1
REFCLK_P_1
HSIP_1
HSIN_1
HSOP_1
HSON_1
U9
GND
CLKREQB
(NC) DVDD33
REFCLK_N
REFCLK_P
(DVDD10) EVDD33
HSIN
HSIP
(DVDD10) DVDD10
HSON
HSOP
(EVDD10) EVDD09
R215 4.7K R0201
R216 4.7K R0201
R281 1K R0201
R280 1K R0201
R74 0R R0201
[11] PCIE20_1_2_WAKEn_M1_L_3V3
ISOLATEB_1
LANWAKEB_1
GPI/GPO_1
VDDREG_1
POW_EXT_SWR_1
R283 4.7K R0201
C315 2.2uF/10V C0402
CKXTAL2_PHY1
CKXTAL1_PHY1
R284 2.49K_1% R0201
RSET_PHY1
ISOLATEB
LANWAKEB
GPI (GPO)
NC/VDDREG (DVDD33)
NC/VDDREG
POW_EXT_SWR/ENSWREG
NC/REG_OUT
DVDD09 (DVDD10)
CKXTAL2
CKXTAL1
AVDD09 (AVDD10)
RSET
RTL8125B(S)
RTL8125BG(S)
RTL8111K(I)
PERSTB
(NC) LED3
(NC) AVDD33_XTAL
(NC) AVDD33
(SMBDATA) EEPRON_SEL/PPS_PIN
EECS/SCL
(EEDI/SI/SDA) LED0/EEDI/SPISI/SDA
DVDD33
(NC) DVDD09
LED1/EESK/SPISCK
LED0/SPICSB) SPICSB
LED2/EEDO/SPISO
PERSTB_PHY1
R450 10K R0201
C121 100nF/10V C0201
R73 0R R0201
PCIE_PERST_L_3V3 [11]
EEPROM_SEL_1
EECS/SCL_1
LED0/EEDI/SPISI/SDA_1
LED1/EESK/SPISCK_1
SPICSB/LED0_1
LED2/EEDO/SPISO_1
R274 4.7K R0201
RTL8125B
QFN48_RTL8125B
MDIP0
MDIN0
AVDD33
AVDD09 (AVDD10)
MDIP1
MDIN1
AVDD33
MDIP2
MDIN2
AVDD09 (AVDD10)
MDIP3
MDIN3
PHY1_MDIP-
PHY1_MDIN-
PHY1_MDI0
PHY1_MDI1
PHY1_MDI2
PHY1_MDI3
AVDD33_PLL1
0V95_PHY1
CKXTAL1_PHY1
Y2
X1
GND
X2
25MHz_12PF
CRY4_3R20X2R50X0R32
CKXTAL2_PHY1
C119 18pF/50V C0201
C120 18pF/50V C0201
TWSI
EEPROM
R309 NC R0201
R442 NC R0201
LED0/EEDI/SPISI/SDA_1
56/66
46
R416 NC R0201
R443 NC R0201
EEPROM_SEL_1
R444 NC R0201
R445 NC R0201
EECS/SCL_1
U23
QG2439S
xfmr-smd_30f-81anl
TX1+
TX1-
TX2+
TX2-
TX3+
TX3-
TX4+
TX4-
TD1+
TD1-
TD2+
TD2-
TD3+
TD3-
TD4+
TD4-
CT3
CT4
CT9
CT10
CT22
21CT
16CT
15CT
C148 100nF/25V C0402
C149 100nF/25V C0402
TC1_1
TC2_1
TC3_1
TC4_1
R177 75R R0402
R176 75R R0402
R175 75R R0402
R178 75R R0402
PHY1_MD3N
PHY1_MD3P
PHY1_MD2N
PHY1_MD2P
PHY1_MD1N
PHY1_MD1P
PHY1_MD0N
PHY1_MD0P
C152 1nF/2KV C1206
RJ45_GND
NC/ESD5341N
ESD0402
ED9
ED10
ED11
ED12
ED13
ED14
ED15
ED16
SPICSB/LED0_1
LED2/EEDO/SPISO_1
R85 NC R0201
R84 0R R0201
LED0/EEDI/SPISI/SDA_1
LED1/EESK/SPISCK_1
R82 NC R0201
R83 0R R0201
R81 0R R0201
CN9
RCHSP32-DW-GY-LB2
RJ45_RCHSP32_DW
C155 1nF/2KV C1206
U26
SY8089AAAC
SOT_23_5
VIN
GND
EN
LX
FB/OUT
C321 10uF/10V C0402
R88 10K R0201
R308 100K R0201
L8
WPN252012H2R2MT
IND_252012
R287 68K_1% R0201
R307 120K_1% R0201
R78 0R R0603
R89 0R R0603
C322 10uF/10V C0402
EXTSWR_CORE
0V95_PHY2
HSIP_1
HSIN_1
HSOP_1
HSON_1
C115 100nF/10V C0201
C116 100nF/10V C0201
C117 100nF/10V C0201
C118 100nF/10V C0201
PCIE20_0_TXP [10]
PCIE20_0_TXN [10]
PCIE20_0_RXP [10]
PCIE20_0_RXN [10]
REFCLKP_1
REFCLKN_1
R71 0R R0201
R72 0R R0201
PCIE20_0_REFCLKP [10]
PCIE20_0_REFCLKN [10]

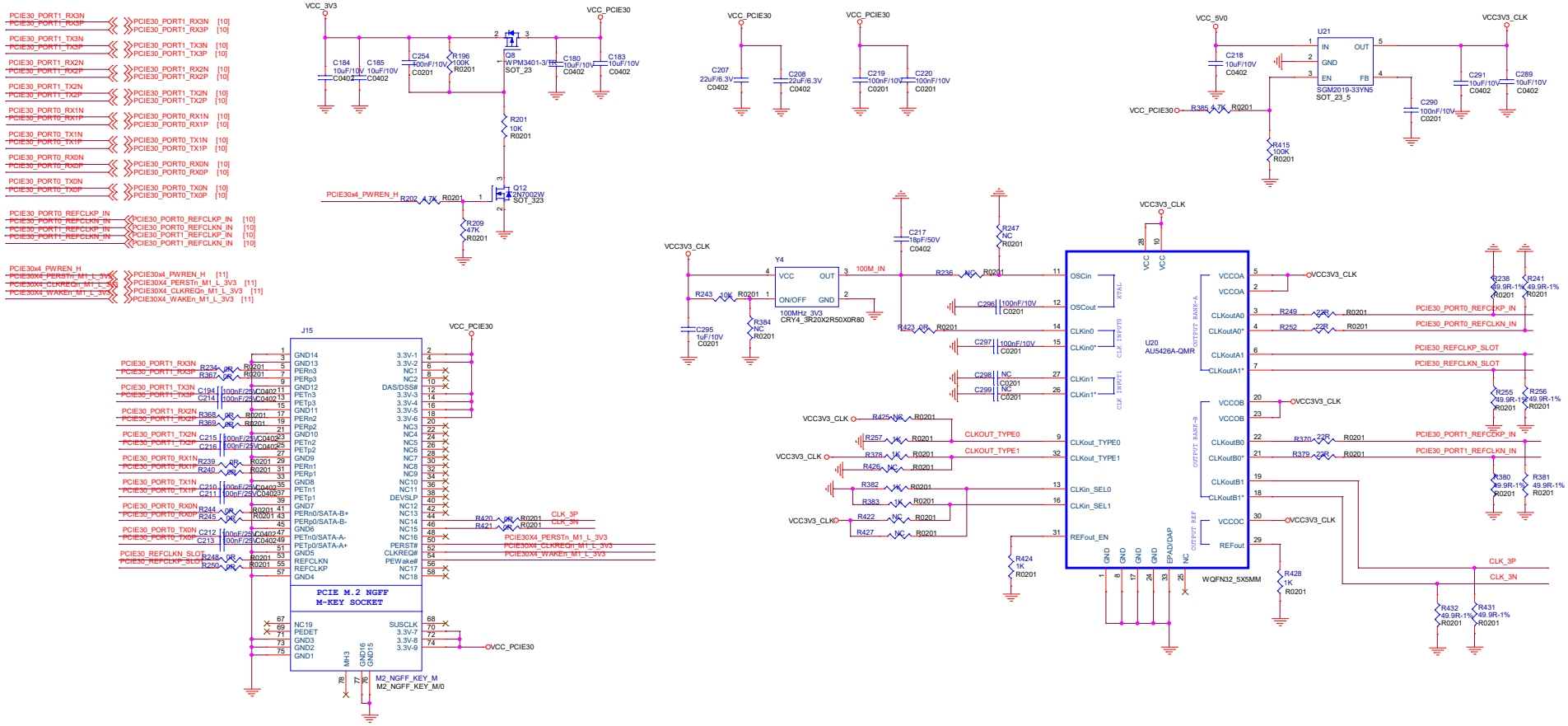
PCIE TO RJ45


3V3_PHY2
VCC_3V3_S0
R30 0R R0603
C21 4.7uF/10V C0402
C22 4.7uF/10V C0402
C23 4.7uF/10V C0402
C24 4.7uF/10V C0402
R31 180R_1A R0603
AVDD33_PLL2
C25 10uF/10V C0402
R32 0R R0402
VDDREG_2
C26 100nF/10V C0201
C27 10uF/10V C0402
0V95_PHY2
C28 4.7uF/10V C0402
C32 4.7uF/10V C0402
C33 10uF/10V C0402
C34 100nF/10V C0201
C75 100nF/10V C0201
R33 NC R0402
EVDD33/DVDD10_PHY2
C76 1uF/10V C0201
R34 0R R0402
3V3_PHY2
R35 NC R0402
GNDTX/DVDD10_PHY2
C77 NC C0201
[11] PCIE30x1_0_CLKREQn_M1_L_3V3
R69 4.7K R0201
R70 0R R0201
CLKREQB_2
R90 4.7K R0201
R91 4.7K R0201
R92 1K R0201
R93 1K R0201
R94 1K R0201
[11] PCIE30x1_0_WAKEn_M1_L_3V3
ISOLATEB_2
LANWAKEB_2
GPI/GPO_2
R95 4.7K R0201
C80 2.2uF/10V C0402
POW_EXT_SWR_2
CKXTAL2_PHY2
CKXTAL1_PHY2
R96 2.49K 1% R0201
RSET_PHY2
U10
RTL8125B(S)
RTL8125BG(S)
RTL8111K(I)
RTL8125B
QFN48_RTL8125B
CKXTAL1_PHY2
Y3
X1 GND
GND X2
25MHz_12PF
CRY4_3R20X2R50X0R80
C78 18pF/50V C0201
C79 18pF/50V C0201
CKXTAL2_PHY2
HSIP_2
HSIN_2
HSOP_2
HSON_2
C133 100nF/10V C0201
C134 100nF/10V C0201
C135 100nF/10V C0201
C136 100nF/10V C0201
PCIE20_1_TXP [10]
PCIE20_1_TXN [10]
PCIE20_1_RXP [10]
PCIE20_1_RXN [10]
REFCLKP_2
REFCLKN_2
R119 0R R0201
R120 0R R0201
PCIE20_1_REFCLKP [10]
PCIE20_1_REFCLKN [10]
PERSTB_PHY2
R98 10K R0201
C137 100nF/10V C0201
R99 0R R0201
PCIE30x1_0_PERSTn_M1_L_3V3 [11]
EEPROM_SEL_2
EECS/SCL_2
LED0/EEDI/SPISI/SDA_2
LED1/EESK/SPISCK_2
SPICSB/LED0_2
LED2/EEDO/SPISO_2
R100 4.7K R0201
TWSI
EEPROM
R101 NC R0201
R102 NC R0201
56/66
46
R105 NC R0201
R118 NC R0201
R103 NC R0201
R104 NC R0201
EECS/SCL_2
U24
QG2439S
xfmr-smd_30f-81anl
PHY2_MDI3-
PHY2_MDI3+
PHY2_MDI2-
PHY2_MDI2+
PHY2_MDI1-
PHY2_MDI1+
PHY2_MDI0-
PHY2_MDI0+
C138 100nF/25V C0402
C139 100nF/25V C0402
TX1+
TX1-
TX2+
TX2-
TX3+
TX3-
TX4+
TX4-
CT3
CT4
CT9
CT10
TD1+
TD1-
TD2+
TD2-
TD3+
TD3-
TD4+
TD4-
CT22
21CT
16CT
15CT
PHY2_MD3N
PHY2_MD3P
PHY2_MD2N
PHY2_MD2P
PHY2_MD1N
PHY2_MD1P
PHY2_MD0N
PHY2_MD0P
TC1_2
TC2_2
TC3_2
TC4_2
R121 75R R0402
R122 75R R0402
R123 75R R0402
R124 75R R0402
C140 1nF/2KV C1206
RJ45_GND
NC/ESD5341N
ESD0402
ED1
ED4
ED5
ED6
ED7
ED8
ED17
ED16
SPICSB/LED0_2
LED2/EEDO/SPISO_2
R139 NC R0201
R140 10R R0201
LED0/EEDI/SPISI/SDA_2
R142 NC R0201
R143 0R R0201
LED1/EESK/SPISCK_2
R141 10R R0201
CN8
RCHSP32-DW-GY-LB2
RJ45_RCHSP32_DW
RJ45_GND

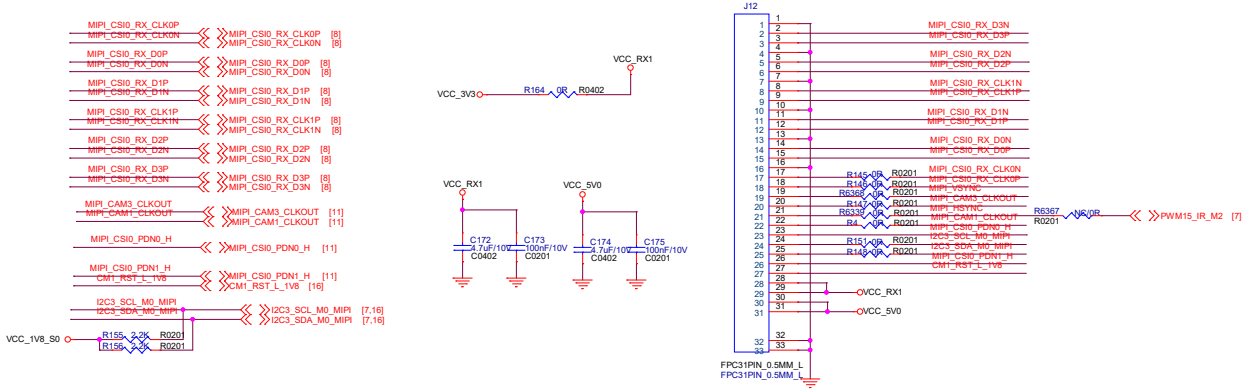
TF CARD



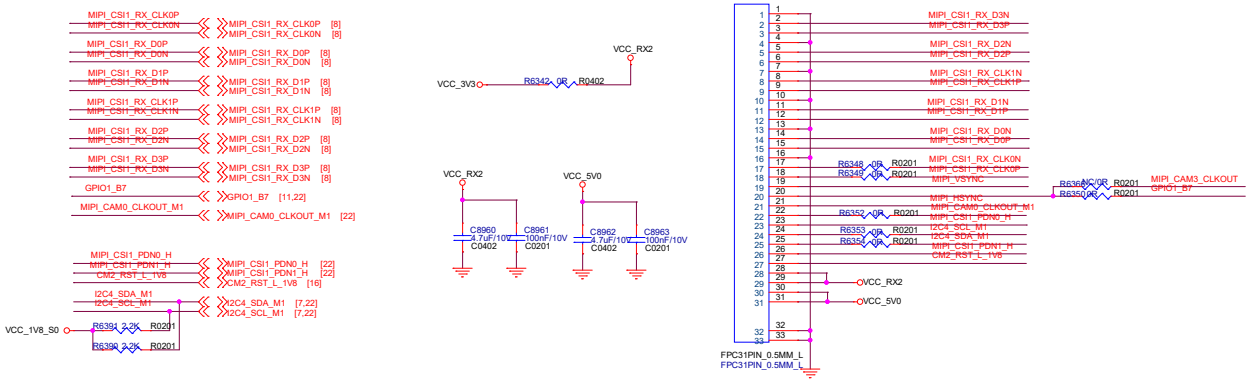
M.2_PCIE3.0



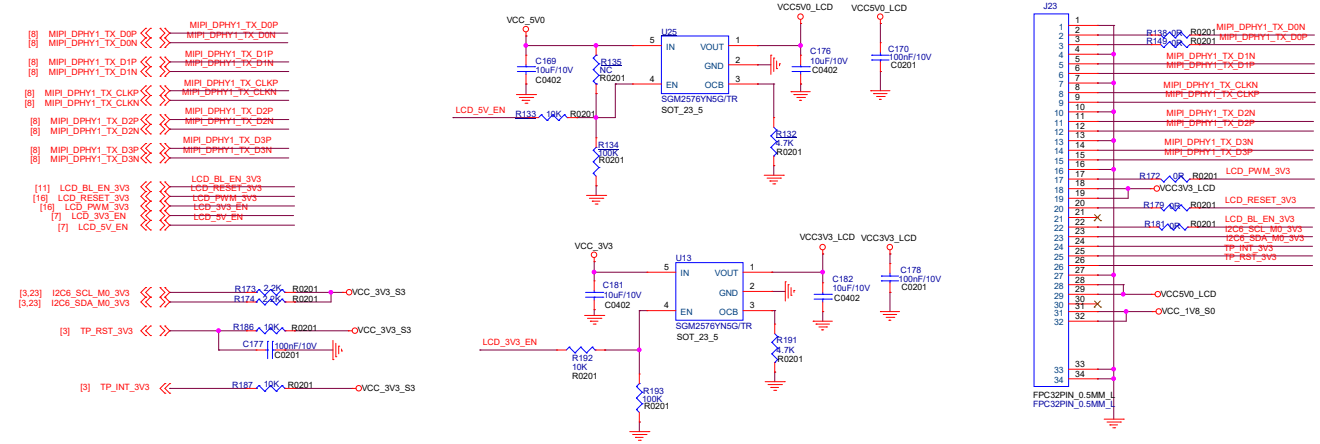
CSI0_MIPI

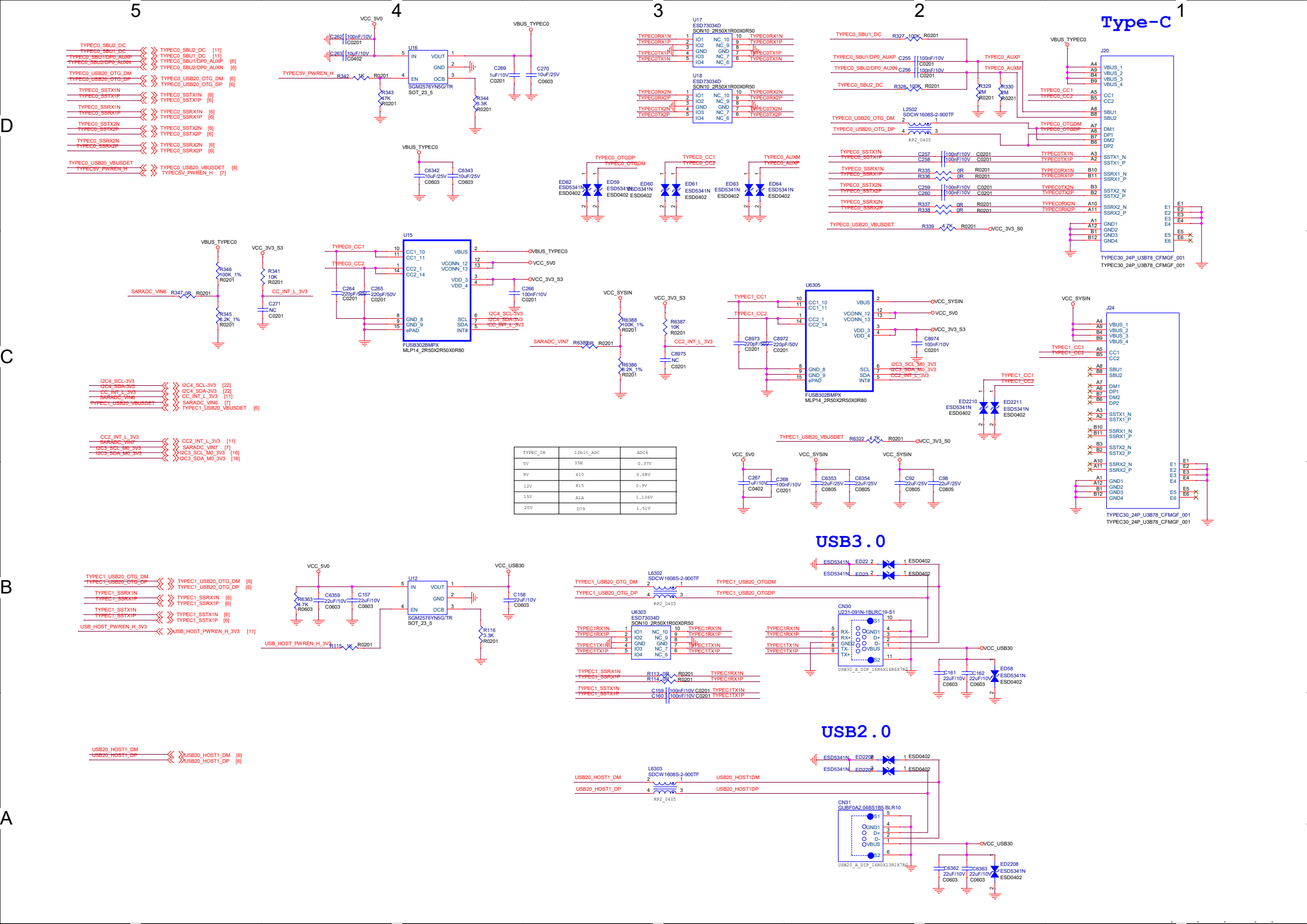


CS1_MIPI



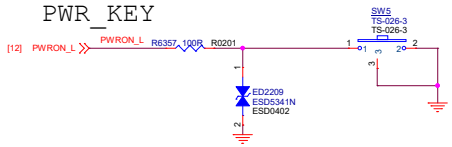
DSI_MIPI



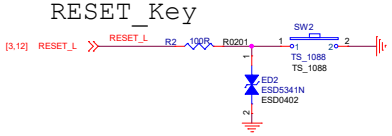

Type-C
USB3.0
USB2.0
TYPEC0_SBU2_DC
TYPEC0_SBU1_DC
TYPEC0_SBU1/DP0_AUXP
TYPEC0_SBU2/DP0_AUXN
TYPEC0_USB20_OTG_DM
TYPEC0_USB20_OTG_DP
TYPEC0_SSTX1N
TYPEC0_SSTX1P
TYPEC0_SSRX1N
TYPEC0_SSRX1P
TYPEC0_SSTX2N
TYPEC0_SSTX2P
TYPEC0_SSRX2N
TYPEC0_SSRX2P
TYPEC0_USB20_VBUSDET
TYPEC5V_PWREN_H
VCC_5V0
VBUS_TYPEC0
VCC_3V3_S3
VCC_SYSIN
VCC_USB30
U16
SGM2576YN5G/TR
SOT_23_5
U17
U18
ESD73034D
SON10_2R50X1R00X0R50
U15
FUSB302BMPX
MLP14_2R50X2R50X0R80
U6305
J20
J24
TYPEC30_24P_U3B78_CFMGF_001
L2502
SDCW1608S-2-900TF
L6302
L6303
RP2_0405
CN30
CN31
U231-091N-1BLRC19-S1
GUBF0A2.04BS1B5.BLR10
USB30_A_DIP_16R6X14R6X7R0
USB20_A_DIP_14R0X13R1X7R0
USB_HOST_PWREN_H_3V3
USB20_HOST1_DM
USB20_HOST1_DP
TYPEC1_USB20_OTG_DM
TYPEC1_USB20_OTG_DP
TYPEC1_SSRX1N
TYPEC1_SSRX1P
TYPEC1_SSTX1N
TYPEC1_SSTX1P
TYPEC1_USB20_VBUSDET
SARADC_VIN6
SARADC_VIN7
I2C4_SCL-3V3
I2C4_SDA-3V3
CC_INT_L_3V3
I2C3_SCL_M0_3V3
I2C3_SDA_M0_3V3
TYPEC_IN
12bit_ADC
ADC6
5V
9V
12V
15V
20V
35E
61O
81S
A1A
D79
0.37V
0.68V
0.9V
1.136V
1.51V

KEY

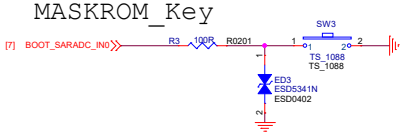
PWR_KEY



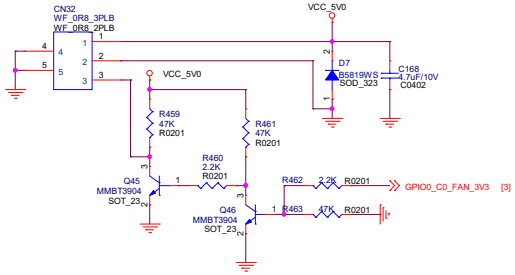
RESET_Key

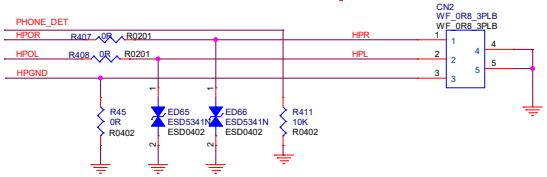


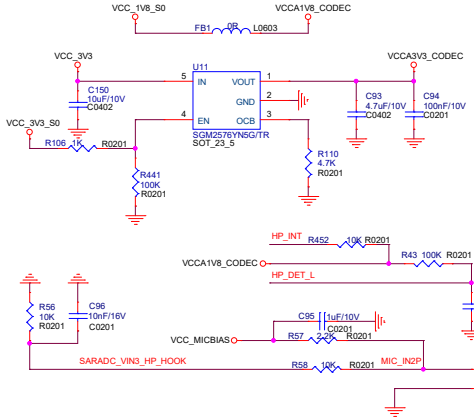
MASKROM_Key

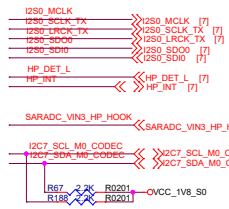


FAN

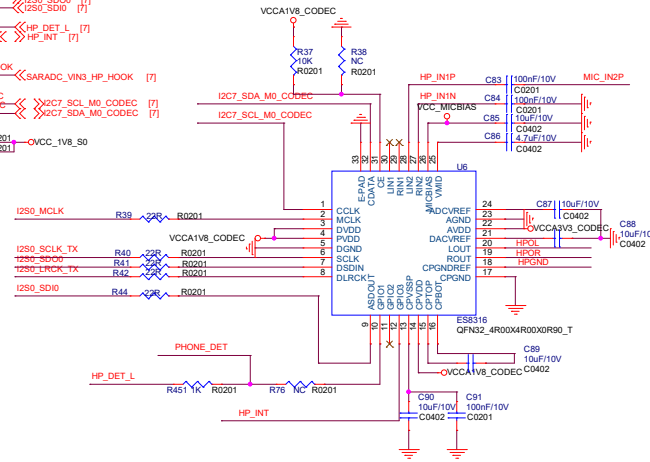


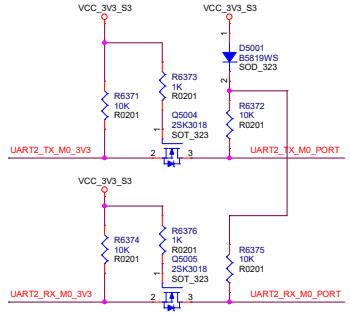




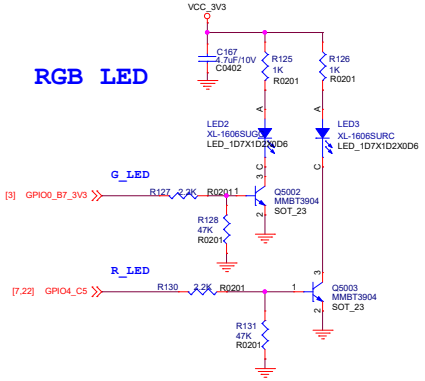


CODEC ES8316

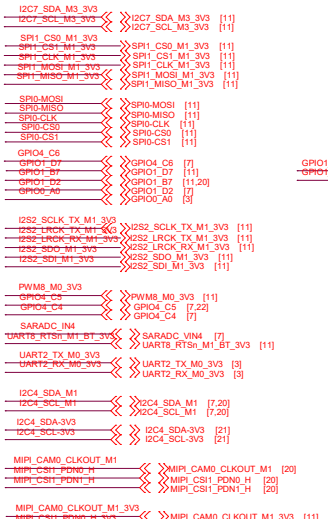


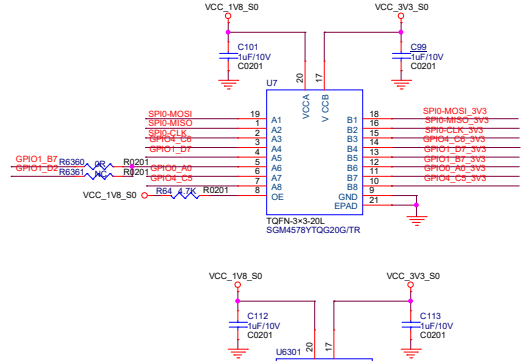


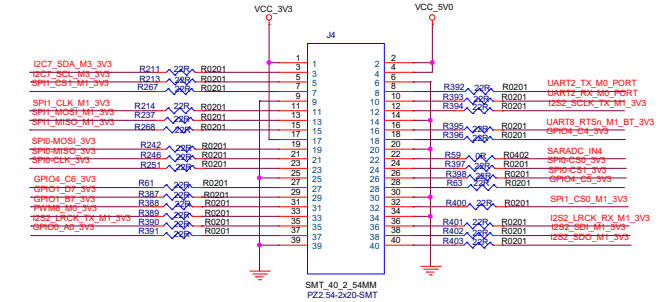
RGB LED

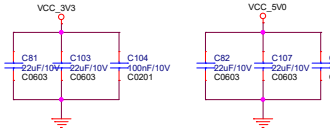


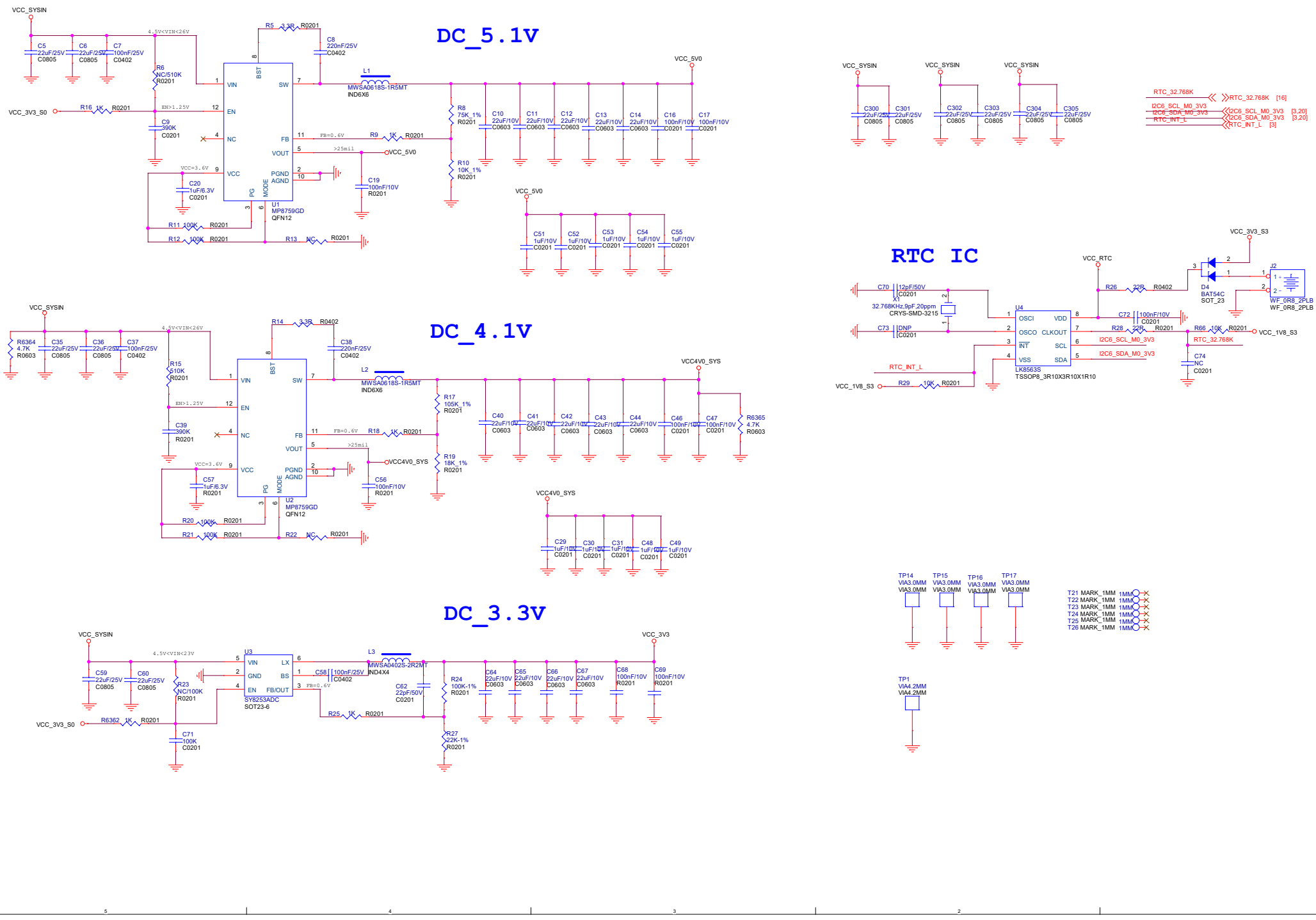
40PIN_GPIO










DC_5.1V
VCC_SYSIN
4.5V<VIN<26V
C5 22uF/25V C0805
C6 22uF/25V C0805
C7 100nF/25V C0402
R5 3.3R R0201
C8 220nF/25V C0402
R6 NC/510K R0201
L1 MWSA0618S-1R5MT IND6X6
VCC_5V0
VCC_3V3_S0
R16 1K R0201
EN>1.25V
C9 390K C0201
R8 75K_1% R0201
C10 22uF/10V C0603
C11 22uF/10V C0603
C12 22uF/10V C0603
C13 22uF/10V C0603
C14 22uF/10V C0603
C16 100nF/10V C0201
C17 100nF/10V C0201
R10 10K_1% R0201
FB=0.6V
R9 1K R0201
>25mil
VCC=3.6V
C20 1uF/6.3V C0201
C19 100nF/10V R0201
U1 MP8759GD QFN12
BST SW VIN EN NC FB VOUT PGND AGND VCC PG MODE
R11 100K R0201
R12 100K R0201
R13 NC R0201
C51 1uF/10V C0201
C52 1uF/10V C0201
C53 1uF/10V C0201
C54 1uF/10V C0201
C55 1uF/10V C0201
C300 22uF/25V C0805
C301 22uF/25V C0805
C302 22uF/25V C0805
C303 22uF/25V C0805
C304 22uF/25V C0805
C305 22uF/25V C0805
RTC_32.768K
RTC_32.768K [16]
I2C6_SCL_M0_3V3
I2C6_SDA_M0_3V3
RTC_INT_L
I2C6_SCL_M0_3V3 [3,20]
I2C6_SDA_M0_3V3 [3,20]
RTC_INT_L [3]
DC_4.1V
R6364 4.7K R0603
C35 22uF/25V C0805
C36 22uF/25V C0805
C37 100nF/25V C0402
R14 3.3R R0402
C38 220nF/25V C0402
R15 510K R0201
L2 MWSA0618S-1R5MT IND6X6
VCC4V0_SYS
R17 105K_1% R0201
R18 1K R0201
R19 18K_1% R0201
C39 390K R0201
C40 22uF/10V C0603
C41 22uF/10V C0603
C42 22uF/10V C0603
C43 22uF/10V C0603
C44 22uF/10V C0603
C46 100nF/10V C0201
C47 100nF/10V C0201
R6365 4.7K R0603
C57 1uF/6.3V R0201
C56 100nF/10V R0201
U2 MP8759GD QFN12
R20 100K R0201
R21 100K R0201
R22 NC R0201
C29 1uF/10V C0201
C30 1uF/10V C0201
C31 1uF/10V C0201
C48 1uF/10V C0201
C49 1uF/10V C0201
RTC IC
VCC_3V3_S3
VCC_RTC
C70 12pF/50V C0201
X1 32.768KHz,9pF,20ppm CRYS-SMD-3215
C73 DNP C0201
U4 LK8563S TSSOP8_3R10X3R10X1R10
OSCI VDD OSCO CLKOUT INT SCL VSS SDA
R26 22R R0402
C72 100nF/10V C0201
R28 22R R0201
R66 10K R0201
D4 BAT54C SOT_23
J2 WF_0R8_2PLB
VCC_1V8_S3
R29 10K R0201
C74 NC C0201
I2C6_SCL_M0_3V3
I2C6_SDA_M0_3V3
RTC_INT_L
RTC_32.768K
DC_3.3V
VCC_SYSIN
4.5V<VIN<23V
C59 22uF/25V C0805
C60 22uF/25V C0805
U3 SY8253ADC SOT23-6
VIN LX GND BS EN FB/OUT
L3 MWSA0402S-2R2MT IND4X4
VCC_3V3
C58 100nF/25V C0402
FB=0.6V
R23 NC/100K R0201
R25 1K R0201
C62 22pF/50V C0201
R24 100K-1% R0201
C64 22uF/10V C0603
C65 22uF/10V C0603
C66 22uF/10V C0603
C67 22uF/10V C0603
C68 100nF/10V R0201
C69 100nF/10V R0201
R27 22K-1% R0201
VCC_3V3_S0
R6362 1K R0201
C71 100K C0201
TP14 VIA3.0MM
TP15 VIA3.0MM
TP16 VIA3.0MM
TP17 VIA3.0MM
T21 MARK_1MM
T22 MARK_1MM
T23 MARK_1MM
T24 MARK_1MM
T25 MARK_1MM
T26 MARK_1MM
TP1 VIA4.2MM

Revision History

Version	Date	By	Change Dsecription	Approved
V1.0	2023-08-07	SL Chen	First release;	
V1.0	2023-09-19	SL Chen	1.phone jack change to connector; change U23&U24 singal for layout,del poe cn1; 2.PWR: add C8964&C8965,R6364&R6365; 3.CAM: add R6366&R6367&R6368;	
V1.1	2023-12-13	SL Chen	1.VOL_IC: U27/U7/U6301change part pin; 2.USB：U15 from TypeC1 change to TypeC0; CN31 from USB3.0 change to USB2.0; 3.40PIN：add UART Protect; J2&CN2&CN32 change to WF0.8; 4.CAM：DEL R6352, CSI_1 I2C3 change to I2C4 5.WIFI Module: AP6275S change to AP6275P;	